

Solution strategies for the delamination analysis based on a combination of local-control arc-length and line searches

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SUMMARY

The paper describes a local-control arc-length method which can be combined with various forms of line-search procedure. In particular, a new 'double-line-search' method is developed, which significantly improves the solution procedure and turns out to be efficient and robust. Although the potential range of applications is wide, the method is here limited to the finite element analysis of delamination in a laminated composite using a cohesive-zone model combined with interface elements. Three problems have been analysed and comparisons have been made with experimental results. Copyright © 2003 John Wiley & Sons, Ltd.

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1. INTRODUCTION

Cohesive-zone models have extensively been used for the analysis of delamination. The assumption of the existence of a cohesive zone around the crack tip, which separates an undamaged area from a fully delaminated part of an interface, was initially made by Dugdale [1] and Barenblatt [2]. It was then in Reference [3] that, for the first time, this idea was used in a finite-element analysis to develop a 'traction-relative displacement' law for '0-thickness' interface elements, which fulfils the energy balance criterion of fracture mechanics. Numerous contributions have then followed and, among many others, cohesive-zone models were used in References [4–8], for the analysis of toughness in ductile materials, and in References [9–14], for the simulation of crack-growth in composites. Mixed-mode delamination has often been addressed [7, 8, 10, 11, 13, 14].

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In this paper, the interface-element formulation proposed in Reference [15], and later studied in References [16–18], is used for the analysis of delamination within a displacement-based finite-element model. Attention is focused on the numerical solution of the non-linear problem posed by the propagation of the delamination.

The local softening interface relationship used often results in a global softening response of the structure. Several limit points are then typically present in the equilibrium path, if this is determined assuming a quasi-static evolution, that is if the external loads and the prescribed displacements are supposed to continuously adjust themselves so as to be always in equilibrium with the internal forces. Arc-length methods are generally used in these cases.

In ‘real life’, the loads do not adjust themselves and hence, even if they are applied quasi-statically, the limit points induce dynamic phenomena, like ‘dynamic snaps’ from one to another stable branch of the equilibrium path, or the final structural collapse. Their type and position generally depend on the type of loading, but in a finite-element simulation additional limit points may occur as an artifact of the discretization [18].

In many cases the quasi-static solution provides sufficient information to deduce the real response of the structure. In particular, this occurs if the unstable branches of the equilibrium path clearly indicate the starting and the ending point of each dynamic snap, and when the deformation history during the snap does not significantly affect the final results (for example, this is the case for the multiple-delamination problem analysed in Section 4.2 under displacement control). Otherwise, in order to directly predict the structural response, a very convenient strategy is to use algorithms which can switch from statics to implicit or explicit dynamics [19]. Nevertheless, for their success it is still essential that an efficient and robust quasi-static algorithm is used to overcome the limit points in order to detect the beginning of an unstable branch and then switch to dynamics.

It is worth noting that an explicit approach could also be used within a quasi-static analysis and would have the advantage, with respect to implicit methods, that equilibrium does not have to be rigorously satisfied at each increment. However, explicit methods also require very small time steps in order to avoid that the solution diverts from the equilibrium path, which can never occur if an implicit method is used. A comparison between explicit and implicit methods would then be important but this also requires that both approaches are developed at their best.

In this paper we develop efficient, fully implicit quasi-static algorithms for the analysis of delamination. In the lights of the previous considerations, they can be efficiently adopted for many applications and will also represent a useful ‘ingredient’ for further developments of strategies which combine statics and dynamics. The developed algorithms are based on an arc-length method which falls within the family of the so-called ‘local-control’ procedures [9, 10, 20–23]. These are particularly suited to the analysis of a delamination process, in which strong non-linearities are mainly related to a localized set of degrees of freedom. The control function is here defined as a weighted sum of a localized set of relative-displacement parameters and, for each increment, these parameters are selected with a view to making the control function locally increase. The treatment is currently restricted to geometrically linear problems so that a linear constraint equation is obtained.

The numerical performances of the developed local-control procedure have been compared with those of the cylindrical arc-length method, in its modified version presented in Reference [24], which involves a global norm of the displacements. The local-control method is generally more efficient but, for some of the problems analysed, it has been found to be more unstable.

The ‘lack of robustness’ is typically related to the extreme ‘non-smoothness’ of the problem and the use of line searches was considered as a remedy. Although line searches are generally not required for relatively smooth structural problems if the full Newton method is used [25], it has already been shown in Reference [26] how they can play a significant role when the problem is non-smooth.

The term η -line search will be used to indicate the classical procedure whereby an optimal scaling factor, η , for the iterative change of the displacements, $\delta\mathbf{p}$, is sought. When the problem is governed by a potential, and the load is kept fixed within each increment, the η -line search is implemented in order to minimize the potential energy. However, such a nice theoretical framework cannot immediately be applied to a non-conservative problem or in combination with a path-following technique, which allows for a change of the load intensity during each iteration. Nevertheless, on the basis of simple equilibrium considerations, it is still required that the projection $\mathbf{g}(\eta) \cdot \delta\mathbf{p}$ of the residual force vector, $\mathbf{g}$, along the direction of $\delta\mathbf{p}$ be below a given tolerance [26–28].

The strategies for coupling arc-length control and η -line searches depend on whether a linear or a non-linear constraint equation is used. Since the local-control method here studied is based on a linear constraint equation, the approach used in Reference [25], which coincides with that indicated in Reference [29] for the case of a linear constraint, has been followed, although the implementation of the line-search algorithm differs because extrapolations are allowed here, whereas they were not in References [25, 29].

Although the use of η -line searches increased the stability of the solution procedure, the algorithm still encountered severe convergence difficulties for some of the problems analysed, which required a large number of reductions of the increment length. On some occasions, the automatic incrementation procedure failed to find a good increment length for convergence and the analysis had to be terminated. Such difficult steps always coincided with very sharp turns in the equilibrium path. A closer look revealed that such convergence problems generally occur when the η -line search fails to find an optimal step length, η . For such cases, a modification to the solution procedure has been developed. It involves two types of line search and is then indicated as a ‘double-line-search’ method. If the η -line search fails, with the new method a unit step length is assumed ($\eta = 1$), and a new iterative change of the load factor is sought. This new change, which is denoted by $\delta\lambda^*$ and is different from that initially computed from the solution to the arc-length equation, yields a new search direction, $\delta\mathbf{p}^*$, as well as a new residual vector $\mathbf{g}^*$. The new type of line search, indicated as a $\delta\lambda$ -line search, is used to compute $\delta\lambda^*$, with a view to making the product $\mathbf{g}^* \cdot \delta\mathbf{p}^*$ sufficiently small. It is proved that, provided that a sufficiently large number of line-search iterations is used, either convergence is directly achieved for the $\delta\lambda$ -line search, or two values, $\delta\mathbf{p}^*$ and $\delta\lambda^*$, are computed whereby a new η -line search will then converge. Switching on the ‘double-line-search’ method results in a violation of the arc-length constraint. A criterion is then developed to compute a new temporary value of the ‘increment length’, Δl .

Three examples concerning carbon-fiber laminated composites are discussed in the paper and the numerical performances of four types of solution procedure are compared in detail. The first procedure is the cylindrical arc-length (CAL) method, in its version proposed in Reference [24]. The others are the local-control method, developed here, with ‘no line searches’ (NLS method), the local-control method supplemented only with η -line searches, referred to as the ‘single line-search’ (SLS) method and, finally, the local-control with the new ‘double-line-search’ procedure (DLS method).

2. INTERFACE MODEL AND FINITE ELEMENT DISCRETIZATION

Interface delamination may be numerically modeled either by using methods directly based on fracture mechanics or by assuming the existence of a ‘cohesive zone’ around the crack tip, where the tractions along the interface decrease from a finite value to zero (see among others References [3–14]). Hereafter, we will follow the latter approach and use the interface model developed in References [15–18]. Reference [18] gives an extensive discussion of the underlying mechanics.

At each point of the interface a relative-displacement vector, $\mathbf{s}$, is defined, which is null in the initial configuration. A local reference system is pointwise introduced with axis 1 normal to the interface. For a two-dimensional problem, whereby the interface is a line of the initial configuration, axis 2 is directed along the tangent to the interface. For a three-dimensional problem, for which the interface is a surface of the initial configuration, axes 2 and 3 lie in the tangent plane to the interface. Accordingly, the relative displacement can be decomposed into three components along the axes, s_1 , s_2 and s_3 , usually named modes I, II and III.

Input parameters of the model are, for each mode, the values G_{ci} , s_{oi} and t_{oi} , which represent the fracture energy, the ‘cracking’ relative-displacement component in mode i and the peak value of the related traction component. These values enter the single-mode interface laws which are discussed in detail in Reference [18] and are graphically depicted in Figure 1, with Figure 1(a) relating to mode I and Figure 1(b) relating to modes II and III.

Since the areas enclosed by each bilinear law of Figure 1 are equal to the fracture energies, G_{ci} , which can be measured by means of experimental tests on standardized specimens, the ‘critical’ relative-displacement values are then taken equal to $s_{ci} = 2 G_{ci}/t_{oi}$.

The ‘cracking’ relative displacements, s_{oi} , are related to the penalty-stiffness values, $E_i = t_{oi}/s_{oi}$, which are chosen in the range which ensures a good prediction of the undamaged behaviour and avoids numerical ill-conditioning.

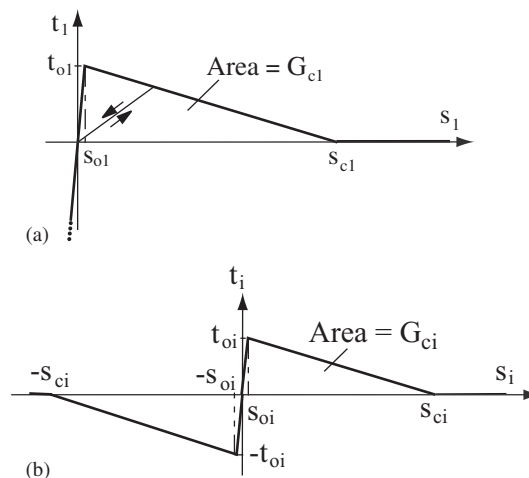


Figure 1. (a) ‘Traction/relative displacement’ relationship for mode I; and (b) ‘traction/relative displacement relationship’ for modes II and III ($i = 2, 3$).

The reliability of the experimental measures for the strengths, t_{oi} , is still an open issue and estimated values are generally given by the experimenters. Such values may have an important influence on the results of the model if initiation of delamination is an important aspect of the simulation or if the material on one or both sides of the interface is ductile [6, 7]. In other cases, relatively wide ranges of strengths produce very similar results for the solution of the boundary-value problem. However, the influence on the finite-element solution is often considerable, as the higher are the strengths, the more refined needs the mesh to be around the process zone in order to avoid spurious and strongly non-smooth oscillations of the computed response [18]. For this reason, analysts often reduce the strengths in order to achieve a smoother, more realistic finite-element solution. However, below certain values this can lead to non-negligible errors, especially in the vicinity of load maxima, which may be underestimated [18]. Other approaches involve adaptive remeshing [30]. However, the authors believe that more promising methods have recently been developed, which are based on the local enrichment of the kinematics in the zone where a crack propagates. Belytschko and his co-workers introduce this idea in the so-called extended finite-element method in conjunction with a pure fracture-mechanics approach, and thus the adopted enrichment involves functions with crack-tip singularities (see, for example, Reference [31]). A related approach has successfully been adopted by the authors in Reference [32] in conjunction with a cohesive-zone model, using one-dimensional beam elements. However, since no-singularity appears in such a context, in this work polynomial hierarchical functions are used to adaptively enrich the finite-element interpolating spaces around the process zone. Here, attention will be mainly focused on the solution procedures, and none of the aforementioned enriched formulations will be considered.

A mixed-mode, elastic-damage interface law will be used, which specializes to the single-mode relationships of Figure 1 when only one of the modes is active. The interaction between the different modes is considered by introducing the following variable, k :

$$k = \left[\left(\frac{\langle s_1 \rangle}{s_{o1}} \right)^\alpha + \left(\frac{|s_2|}{s_{o2}} \right)^\alpha + \left(\frac{|s_3|}{s_{o3}} \right)^\alpha \right]^{1/\alpha} - 1 \quad (1)$$

where the scalar α is a material parameter, the symbol $\langle \cdot \rangle$ denotes the Macaulay brackets, defined by $\langle x \rangle = x$ if $x \geq 0$ and $\langle x \rangle = 0$ otherwise. Furthermore, in order to ensure that the relationship detailed later simultaneously predicts loss of adhesion for all modes, the following assumption is made [18]:

$$\frac{s_{o1}}{s_{c1}} = \frac{s_{o2}}{s_{c2}} = \frac{s_{o3}}{s_{c3}} = 1 - \mu \quad (2)$$

The maximum value, $k_{\max}$, that k has previously attained is introduced as ‘history variable’ and the following relationship is used:

$$\mathbf{t} = \begin{cases} \mathbf{E}\mathbf{s} & \text{if } k_{\max} \leq 0 \\ (\mathbf{I} - d\mathbf{C})\mathbf{E}\mathbf{s} & \text{if } k_{\max} > 0 \end{cases} \quad (3)$$

where $\mathbf{E} = \text{diag}[E_1, E_2, E_3]$, $\mathbf{C} = \text{diag}[h(s_1), 1, 1]$, with $h(s_1)$ defined by

$$h(s_1) = 1 \quad \text{if } s_1 \geq 0 \quad \text{and} \quad h(s_1) = 0 \quad \text{if } s_1 < 0 \quad (4)$$

$\mathbf{I}$ is the identity 3×3 matrix and the damage parameter, d , is introduced as follows:

$$d = \min \left\{ 1, \frac{1}{\mu} \left(\frac{k_{\max}}{1 + k_{\max}} \right) \right\} \quad (5)$$

Notice that the matrix, $\mathbf{C}$, plays a role when $s_1 < 0$, in which case mode I does not interact with the other modes and thus follows the one-dimensional no-penetration relationship.

A point is said to be ‘undamaged’ if $k_{\max} \leq 0$; it is said to be ‘softening’, if $0 < k_{\max} = k < k_c$, where the critical value of k , corresponding to $d = 1$, is:

$$k_c = \frac{\mu}{1 - \mu} \quad (6)$$

a point is then said to be ‘unloading’ if $0 < k_{\max} < k_c$ and $k < k_{\max}$ and finally it is denoted as ‘fully cracked’ if $k_{\max} \geq k_c$. No plasticity is introduced and, in the case of unloading, the interface law follows the secant line to the origin.

It can be proved [18] that, for a fixed mode ratio, the following fracture criterion is fulfilled:

$$\left(\frac{G_1}{G_{c1}} \right)^{\alpha/2} + \left(\frac{G_2}{G_{c2}} \right)^{\alpha/2} + \left(\frac{G_3}{G_{c3}} \right)^{\alpha/2} = 1 \quad \text{with } G_i = \int_0^{s_{ci}} t_i ds_i \quad (7)$$

For $\alpha = 2$ and $\alpha = 4$, the linear criterion proposed in Reference [33] and the ellipse criterion given in Reference [34] are recovered, respectively.

Differentiating the relationships above yields, after some manipulations, the following expression for the material tangent stiffness operator, $\mathbf{E}_t$:

$$\mathbf{E}_t = \begin{cases} \mathbf{E} & \text{if } k_{\max} \leq 0 \\ (\mathbf{I} - d\mathbf{C})\mathbf{E} - \frac{\mathbf{C}\mathbf{E}(\mathbf{s} \otimes \tilde{\mathbf{s}}_{\alpha-1})\mathbf{A}}{\mu(1+k)^{1+\alpha}} & \text{if } 0 < k = k_{\max} < k_c \\ (\mathbf{I} - d\mathbf{C})\mathbf{E} & \text{if } 0 < k_{\max} < k_c \text{ and } k < k_{\max} \\ (\mathbf{I} - \mathbf{C})\mathbf{E} & \text{if } k_{\max} \geq k_c \end{cases} \quad (8)$$

where $\mathbf{A} = \text{diag}[1/s_{o1}^\alpha, 1/s_{o2}^\alpha, 1/s_{o3}^\alpha]$ and, denoting by $\text{sgn}(s_i)$ the sign of s_i , $\tilde{\mathbf{s}}_{\alpha-1}$ is defined by

$$\tilde{\mathbf{s}}_{\alpha-1}^T = [h(s_1)\langle s_1 \rangle^{\alpha-1}, \text{sgn}(s_2)|s_2|^{\alpha-1}, \text{sgn}(s_3)|s_3|^{\alpha-1}] \quad (9)$$

In general, $\mathbf{E}_t$ turns out to be non-symmetric unless $\alpha = 2$ and the material properties of the interface are the same for all of the modes.

2.1. Interface elements

The implementation of the interface model in a finite-element analysis is based on the introduction of interface elements, whose topology is defined by a set of couples of nodes. The nodes of each couple share the same position in the initial configuration but are attached to the two different parts of the interface. They will then occupy two different points in the deformed configuration so as to define the nodal relative displacement. The interface elements have then a null thickness before deformation takes place and their geometry is a line for

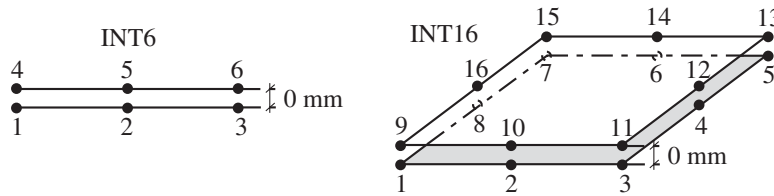


Figure 2. Initial geometries of the 6-noded (INT6) and of the 16-noded (INT16) interface elements.

two-dimensional problems and a surface for three-dimensional ones. In Figure 2 the geometries of the 6-noded element (INT6), used for two-dimensional problems, and of the 16-noded element (INT16), used for three-dimensional problems, are described. They were used for the numerical simulations reported in Section 4. Classic shape functions are used to interpolate the relative displacement, $\mathbf{s}$, from the nodal relative displacements.

In Reference [35] geometric non-linearity was introduced in the formulation via a corotational approach. However, for reasons which will be clarified later, because they are related to the solution procedure used, the treatment will be restricted here to geometrically linear problems. Accordingly, denoting by $\mathbf{p}_e$ the element nodal displacements, a linear ‘relative displacements/nodal displacements’ matrix, $\mathbf{B}_e$, is defined, whereby $\mathbf{s} = \mathbf{B}_e \mathbf{p}_e$. The evaluation of $\mathbf{B}_e$ is straightforward and is not detailed here. The vector of the element internal forces, $\mathbf{q}_{i,e}$, and the element tangent stiffness matrix, $\mathbf{K}_{t,e}$, follow as usual:

$$\mathbf{q}_{i,e} = \int_{\Omega_e} \mathbf{B}_e^T \mathbf{t} d\Omega_e \quad \mathbf{K}_{t,e} = \int_{\Omega_e} \mathbf{B}_e^T \mathbf{E}_t \mathbf{B}_e d\Omega_e \quad (10)$$

where Ω_e denotes the element domain and the traction, $\mathbf{t}$, and the material stiffness, $\mathbf{E}_t$, are computed as previously detailed. A Newton–Cotes integration rule is generally recommended to avoid spurious oscillations of the tractions, and 3 and 9 integration points are used for the INT6 and INT16 elements, respectively.

In order to handle the possibility that the opposite sides of initially already delaminated parts of the interface (initial cracks) come into contact, special contact-type elements INT6C and INT16C are used. Their geometry is the same as for the INT6 and INT16 elements, but the interface law is replaced with the following frictionless contact-type relationship:

$$t_1 = \begin{cases} 0 & \text{if } s_1 \geq 0 \\ E_1 s_1 & \text{if } s_1 < 0 \end{cases} \quad t_2 = t_3 = 0 \quad (11)$$

3. SOLUTION PROCEDURE

We will consider a structure, with one or more embedded interfaces, subjected to a set of forces and/or prescribed displacements, which are supposed to be applied quasi-statically and to depend upon one single load factor, λ . With a standard displacement-based finite-element approximation, the state of the structure depends on an N -dimensional displacement vector, $\mathbf{p}$, with N being the number of the free degrees of freedom, and on the internal variables,

$k_{\max}$, at the integration points of the interface elements. Apart from the non-linear interface law discussed in the previous section, the structure is supposed to be linearly elastic.

The non-linear incremental problem consists of following the structural response for λ varying in a certain range, generally starting from a null initial value and evolving, at least initially, in the direction $\lambda > 0$. The presence of limit points is typical of the response of a structure during delamination and, as is well known, conventional incremental algorithms, which keep the load factor fixed at each increment, fail to overcome such points. Hence, path-following techniques need to be used, whereby the load factor, λ , is added to the set of unknowns. The process is then divided into a finite number of increments. For the n th increment the initial values, $\mathbf{p}_{n-1}$ and λ_{n-1} , are those obtained after convergence at the previous increment, $n - 1$, and therefore are known, together with the previous converged values of the history variables. The final values of the displacements and of the load factor, $\mathbf{p}_n$ and λ_n , are the unknowns, whereas the history variables can be considered as known functions of the displacements through the interface relationship discussed in Section 2. Accordingly the number of unknowns is $N + 1$ and is matched by the N equilibrium equations plus an additional ‘constraint equation’ which accounts for the incrementation of the load factor.

We will consider an arc-length type method, fully detailed later, which is based on the introduction of a ‘control function’ for each increment n , a_n , which depends only on the displacements, $\mathbf{p}$, and somehow gives a measure of the evolution of the structure. The constraint equation is then written as follows:

$$\Delta a_n(\Delta \mathbf{p}_n) = a_n(\mathbf{p}_n) - a_n(\mathbf{p}_{n-1}) = \Delta l_n \quad (12)$$

where $\Delta \mathbf{p}_n = \mathbf{p}_n - \mathbf{p}_{n-1}$ and Δl_n , which represents the increment length, is computed via an automatic incrementation procedure described later. In particular, the control function is made independent from the load factor λ as was suggested by the second author in Reference [36] and by Ramm [37].

Coupling Equation (12) with the equilibrium equations leads to the following non-linear system:

$$\mathbf{g}(\mathbf{p}_n, \lambda_n) = \mathbf{q}_i(\mathbf{p}_n, \lambda_n) - \mathbf{q}_{\text{ext}}(\lambda_n) = \mathbf{0} \quad (13a)$$

$$\Delta a_n(\Delta \mathbf{p}_n) - \Delta l_n = 0 \quad (13b)$$

where $\mathbf{q}_i$ and $\mathbf{q}_{\text{ext}}$ are the vectors of the internal and external forces, respectively, and $\mathbf{g}$ is the residual-force vector. Notice that $\mathbf{q}_i$ is dependent on λ if some displacements are prescribed.

Equations (13) are typically solved by an iterative algorithm. In each iteration, for simplicity of notation, we will generally not use any index related to the iteration number. Instead, we will denote by $\mathbf{p}_o$ and λ_o the values of the displacements and of the load factor computed at the end of the previous iteration, by $\mathbf{p}$ and λ the new values computed at the end of the iteration and by $\delta \mathbf{p}$ and $\delta \lambda$ the iterative changes. Initially, the following simple update is considered:

$$\mathbf{p} = \mathbf{p}_o + \delta \mathbf{p} \quad \lambda = \lambda_o + \delta \lambda \quad (14)$$

Other update formulae will be given later when line searches are considered.

The incremental changes at the end of the iteration are denoted by $\Delta \mathbf{p}$ and $\Delta \lambda$, with $\Delta \mathbf{p} = \mathbf{p} - \mathbf{p}_{n-1}$ and $\Delta \lambda = \lambda - \lambda_{n-1}$. Convergence is achieved at the end of the iteration when:

$$\frac{\|\mathbf{g}(\mathbf{p}, \lambda)\|}{\|\mathbf{f}(\mathbf{p}, \lambda)\|} < \gamma \quad (15)$$

where $\mathbf{f}$ is the vector of all of the external forces, including the reactions, and γ is a numerical tolerance. If convergence is achieved, we set $\mathbf{p}_n = \mathbf{p}$, $\lambda_n = \lambda$ and step forward to the new increment, $n + 1$, otherwise we set $\mathbf{p}_0 = \mathbf{p}$, $\lambda_0 = \lambda$ and iterate. At the first iteration (predictor) we set $\mathbf{p}_0 = \mathbf{p}_{n-1}$ and $\lambda_0 = \lambda_{n-1}$.

The full Newton method is used here so that the equilibrium equations are linearized around $(\mathbf{p}_0, \lambda_0)$, and the following linear system of N equations in $N + 1$ unknowns is obtained:

$$\mathbf{K}_{\text{to}} \delta \mathbf{p} = -\mathbf{g}_0 + \delta \lambda \mathbf{q}'_0 \quad (16)$$

where:

$$\mathbf{K}_{\text{to}} = \left. \frac{\partial \mathbf{q}_i}{\partial \mathbf{p}} \right|_{(\mathbf{p}_0, \lambda_0)} \quad \mathbf{q}'_0 = \left. \frac{\partial \mathbf{q}_{\text{ext}}}{\partial \lambda} \right|_{\lambda_0} - \left. \frac{\partial \mathbf{q}_i}{\partial \lambda} \right|_{(\mathbf{p}_0, \lambda_0)} \quad \mathbf{g}_0 = \mathbf{q}_i(\mathbf{p}_0, \lambda_0) - \mathbf{q}_{\text{ext}}(\lambda_0) \quad (17)$$

Accordingly, $\delta \mathbf{p}$ can be written as follows:

$$\delta \mathbf{p} = \delta \bar{\mathbf{p}} + \delta \lambda \delta \mathbf{p}_t \quad (18)$$

where $\delta \bar{\mathbf{p}}$ and $\delta \mathbf{p}_t$ are given by:

$$\delta \bar{\mathbf{p}} = -\mathbf{K}_{\text{to}}^{-1} \mathbf{g}_0 \quad \delta \mathbf{p}_t = \mathbf{K}_{\text{to}}^{-1} \mathbf{q}'_0 \quad (19)$$

The tangent stiffness matrix, $\mathbf{K}_{\text{to}}$, is not necessarily positive definite, but we assume that it is non-singular, as was always the case in all of our numerical simulations.

By replacing $\Delta \mathbf{p}_n$ in Equation (13b) with the incremental change, $\Delta \mathbf{p} = \mathbf{p}_0 + \delta \bar{\mathbf{p}} + \delta \lambda \delta \mathbf{p}_t - \mathbf{p}_{n-1}$, one equation in the one unknown, $\delta \lambda$, is obtained:

$$\Delta a_n(\Delta \mathbf{p}_0 + \delta \bar{\mathbf{p}} + \delta \lambda \delta \mathbf{p}_t) = \Delta l_n \quad (20)$$

with the vector, $\Delta \mathbf{p}_0 = \mathbf{p}_0 - \mathbf{p}_{n-1}$, being the incremental change of the displacements at the beginning of the iteration. Solving Equation (20) yields, together with relationships (18) and (19), the iterative changes, $\delta \mathbf{p}$ and $\delta \lambda$.

In order to account for the different degrees of convergence difficulties met during the analysis, an automatic incrementation procedure is used. If convergence was reached at increment $n - 1$ with a number I_c of iterations and with the increment length equal to Δl_{n-1} , the new increment length, Δl_n , is evaluated with the formula:

$$\Delta l_n = \left(\frac{I_d}{I_c} \right)^\beta \Delta l_{n-1} \quad (21)$$

where I_d is a user-defined number of 'desired iterations'. The exponent, $\beta = 0.5$, has been used in the applications. If convergence is not achieved in a maximum number of iterations, $N_{\text{it}}^{\text{max}}$, the analysis is restarted with a reduced increment length, $\Delta l_{n,\text{new}} = \beta_{\text{sc}} \Delta l_{n,\text{old}}$. After that a maximum number, $N_{\text{str}}^{\text{max}}$, of consecutive 'increment cuts' has been reached, a final attempt is made with a value, Δl_n , amplified with a factor β_{amp} with respect to the initial value adopted

in the increment. The factors $\beta_{sc} = 0.5$ and $\beta_{amp} = 2.0$ were used in the applications. The first increment size, Δl_1 , is assigned at the beginning of the analysis.

3.1. Local-control arc length and the NLS method

For problems involving softening stress–strain laws or for problems of delamination, expressions of the control function which involve all of the degrees of freedom are not recommended. In fact, sharp snap-backs and snap-throughs are typically related to the localization of the strains, in the case of softening stress–strain laws, and to the increase of debonding in the process zone, in the case of delamination. In both cases, the number of degrees of freedom related to these phenomena are comparatively small. Control functions based on ‘global norms’ of the displacements may turn out to be not sensitive enough and, therefore, ‘local-control’ functions, which only depend on a localized set of degrees of freedom, have recently been used.

For the analysis of strain-softening materials, in Reference [20] de Borst modified the updated normal plane algorithm [37] by replacing the incremental change of the displacements with the crack-mouth sliding displacement or, more generally, by applying weights to the different degrees of freedom. A similar method has been proposed by May and Duan in Reference [22], in which the control function depends on some relative displacements computed within those elements where extensive non-linearity is occurring. Local-control methods have also been used in References [9, 10, 14] for problems involving delamination. One main feature of these strategies is that the control function generally changes from one increment to the next. They can thus be viewed as applications of the ‘subplane control’ method, proposed by Geers in Reference [23], whereby a set of scalar functions, a_i , is defined, among which to choose an optimal control function for each increment of the analysis.

Here, a linear local-control function, a_n , is chosen at each increment, n . Although a_n depends on a localized set of degrees of freedom, as will be shown later in detail, it is formally convenient to make it depend on all of the displacements, $\mathbf{p}$, so that the constraint equation (20) can be given the following expression:

$$\Delta a_n(\Delta \mathbf{p}) = \mathbf{a}_n \cdot (\Delta \mathbf{p}_0 + \delta \bar{\mathbf{p}} + \delta \lambda \delta \mathbf{p}_t) = \Delta l_n \quad (22)$$

with $\mathbf{a}_n$ being an N -dimensional vector. At the first ‘predictor’ iteration $\Delta \mathbf{p}_0 = \mathbf{0}$, because $\mathbf{p}_0 = \mathbf{p}_{n-1}$, and $\delta \bar{\mathbf{p}} = -\mathbf{K}_{10}^{-1} \mathbf{g}_0 \cong \mathbf{0}$, because $\mathbf{g}_0 = \mathbf{g}(\mathbf{p}_{n-1}, \lambda_{n-1}) \cong \mathbf{0}$. At the next ‘corrector’ iterations we can assume that Equation (22) was exactly satisfied at the end of the previous iteration, so that $\Delta l_n = \mathbf{a}_n \cdot \Delta \mathbf{p}_0$. Hence, Equation (22) has one solution given by

$$\delta \lambda \cong \frac{\Delta l_n}{\mathbf{a}_n \cdot \delta \mathbf{p}_t} \quad \text{at the predictor,} \quad \delta \lambda = - \frac{\mathbf{a}_n \cdot \delta \bar{\mathbf{p}}}{\mathbf{a}_n \cdot \delta \mathbf{p}_t} \quad \text{at the corrector} \quad (23)$$

It is often convenient to input an upper bound, $\Delta \lambda_{\max}$, for the magnitude of $\delta \lambda$ at the predictor. In these cases, Δl_n is taken as the minimum value between $\Delta l_{\max} = \Delta \lambda_{\max} |\mathbf{a}_n \cdot \delta \mathbf{p}_t|$ and the value given by Equation (21).

Remark 3.1

A different method would stem from using the following equation:

$$\Delta a_n(\Delta \mathbf{p}) = |\mathbf{a}_n \cdot \Delta \mathbf{p}| = \Delta l_n \quad (24)$$

which would be equivalent to changing the control function and making it non-linear. Two roots, $\delta\lambda_1$ and $\delta\lambda_2$, would then be computed at each iteration:

$$\begin{aligned}\delta\lambda_{1,2} &\cong \pm \frac{\Delta I_n}{\mathbf{a}_n \cdot \delta \mathbf{p}_t} \quad \text{at the predictor} \\ \delta\lambda_{1,2} &= \frac{\pm \Delta I_n - \mathbf{a}_n \cdot (\Delta \mathbf{p}_0 + \delta \mathbf{p})}{\mathbf{a}_n \cdot \delta \mathbf{p}_t} \quad \text{at the corrector}\end{aligned}\quad (25)$$

Also for the spherical and the cylindrical arc-length methods, two exact solutions to the constraint equation can be computed [36] and several methods have been proposed to choose one of them (see for example References [23, 27, 36, 38] for the method to use at the predictor, and References [24, 28, 36] for the choice at the corrector). The same methods can be applied to determine the optimal $\delta\lambda$ in (25).

The two-root issue is a drawback of this alternative approach, but a danger is also hidden in the method which is used here, which leads to Equation (22), because there might be no points, $\mathbf{p}_n$, of the equilibrium path such that $\mathbf{a}_n \cdot \Delta \mathbf{p}_n = \Delta I_n$, or such points might lie too far away to be caught by the solution procedure in one increment, and in either case the algorithm would fail to converge. This problem may still be encountered even after reducing ΔI_n to very small values.

For example, Figure 3(a) shows a (hypothetical) equilibrium path for a two-dimensional problem. It relates to an increment in which there are no intersections, at least in the vicinity of the initial point, $\mathbf{p}_{n-1}$, between the solution path and the straight-line defined by Equation (22). Conversely, Equation (24) defines two lines, and one of them has two intersections with the solution path. In this case relationships (25) would thus be preferable, provided that the procedure used to choose $\delta\lambda$ at each iteration avoids 'doubling back', which in relation to Figure 3(a) would mean the ability to converge to $\mathbf{p}_b$ rather than to $\mathbf{p}_a$. For the same case, the value of the control function is plotted in Figure 3(b) against the length of the equilibrium path, ξ , defined by

$$\xi(\mathbf{p}) = \int_{(0)}^{(\mathbf{p})} \|\delta \pi\| \quad (26)$$

Figure 3(b) shows that the initial point is close to a limit point, ξ_{lim} , for the control function.

Figure 4(a) relates to a case in which relationships (23) are preferable, as there is one intersection between the equilibrium path and the straight line defined by (22). Figure 4(b) indicates that in this case there are no limit points for the control function, which is increasing with respect to the length parameter, ξ .

Summarizing, Equation (22) seems to be preferable to Equation (24) if it is possible to find, for each increment, a control function which is locally monotonically increasing. Then, the solution to the corresponding constraint equation automatically provides both the magnitude and the sign of $\delta\lambda$ at each iteration. Chen and Shreyer in Reference [21] showed that, for structural problems in which material softening occurs, it is possible to choose as a control function some scalar measure of strain which is monotonically increasing. Problems involving delamination are a similar case because, as the delamination propagates in one increment, there is always at least one point of the process zone in which at least one component of the relative displacement is increasing, which would then be a feasible control function. For

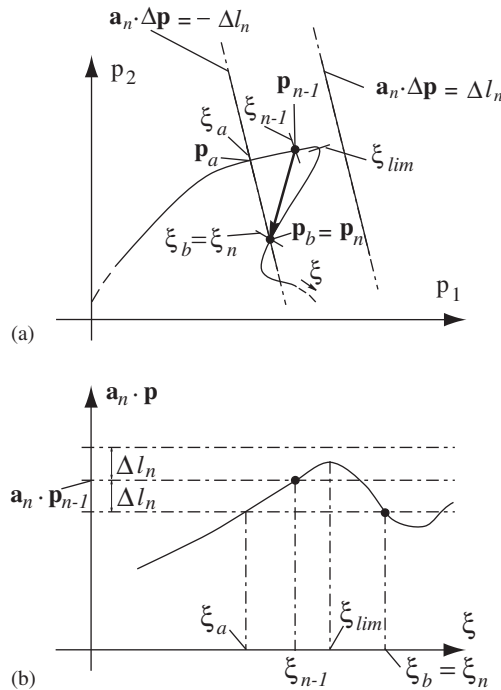


Figure 3. (a) Intersections between the two lines, $\mathbf{a}_n \cdot \Delta \mathbf{p} = \pm \Delta l_n$, and the equilibrium path; and (b) control function against the length, ξ .

problems of multiple delamination, whereby two or more cracks propagate at the same time, and for three-dimensional analyses, in which the crack tip is in fact a line, it is often more effective to use control functions which are weighted sums of a set of relative-displacement components, in such a way that, within each increment, the control function is monotonically increasing.

In order to detail how the vector, $\mathbf{a}_n$, of Equation (23) is built up let us denote by N_i the total number of integration points on all of the interface elements, excluding the contact-type elements. As usual, for the k th integration point, the relative displacement, $\mathbf{s}^k$, is decomposed into its components, or 'modes', s_1^k , s_2^k and, for 3D problems, s_3^k , with $k = 1, \dots, N_i$. The relative-displacement components are then, in total, $2N_i$ for a 2D case and $3N_i$ for a 3D case and they can be arranged in one single array, denoted by δ , so that for a 3D case it is:

$$\delta^T = [s_1^1 \ s_2^1 \ s_3^1 \ s_1^2 \ s_2^2 \ s_3^2 \ \dots \ s_1^{N_i} \ s_2^{N_i} \ s_3^{N_i}] \quad (27)$$

In this way, one single index, i , can be used to point to any one of the relative-displacement components, regardless of what integration point and of what mode it relates to. A critical value, $s_{c,i}^k$, is associated with each component, in accordance with the one-dimensional relationships of Section 2. All the critical values are also ordered in one single vector, δ_c :

$$\delta_c^T = [s_{c1}^1 \ s_{c2}^1 \ s_{c3}^1 \ s_{c1}^2 \ s_{c2}^2 \ s_{c3}^2 \ \dots \ s_{c1}^{N_i} \ s_{c2}^{N_i} \ s_{c3}^{N_i}] \quad (28)$$

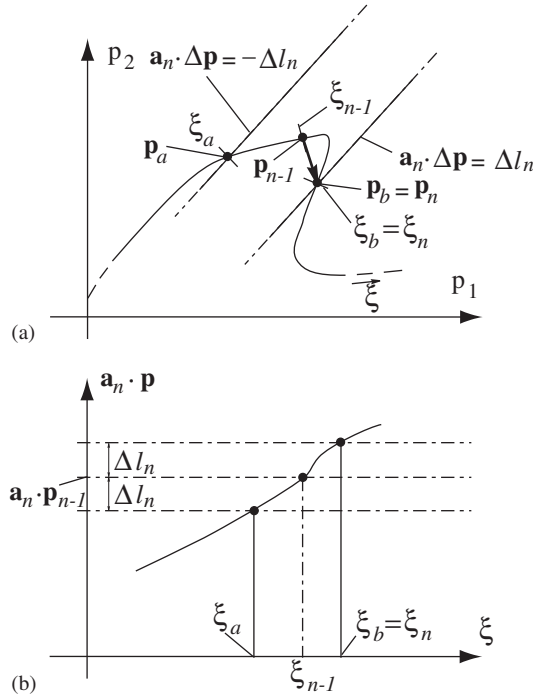


Figure 4. (a) Intersections between the two lines, $\mathbf{a}_n \cdot \Delta \mathbf{p} = \pm \Delta l_n$, and the equilibrium path; and (b) control function against the length, ξ .

In analogy with the notation used for the displacements, let us then denote by δ^m the converged values of the relative-displacement components at the end of the generic increment, m , and simply by δ the values at the end of the iteration under examination. They are linearly linked to the total displacements, $\mathbf{p}_m$ and $\mathbf{p}$, via the relationships:

$$\delta^m = \mathbf{B} \mathbf{p}_m \quad \delta = \mathbf{B} \mathbf{p} \quad (29)$$

with $\mathbf{B}$ being assembled from the element-matrices, $\mathbf{B}_e$, introduced in Section 2.1.

A maximum number, $N_p^{\max}$, of parameters to be controlled is chosen and is here kept fixed for the whole analysis. The actual number of controlled parameters, N_p , with $N_p \leq N_p^{\max}$, is determined within each increment but generally changes from increment to increment. The basic idea is to restrict the attention only to the points within the process zone, and to choose the components which, in the previous converged increment, were the ‘most rapidly changing in the direction of increasing delamination’.

Let us then denote by C_n the set of the indices corresponding to the N_p controlled parameters, which is determined using the following procedure:

1. a mode-I component which was negative at the end of the previous increment is never considered;
2. only the components, δ_i , whose corresponding integraton point was ‘softening’ at the end of the previous increment, are considered;

3. the 'shortlisted' components, δ_i , are then sorted by descending values of their 'normalized incremental change in the direction of increasing delamination', denoted by $\Delta\hat{\delta}_i^{n-1}$ and defined by

$$\Delta\hat{\delta}_i^{n-1} = \text{sgn}(\delta_i^{n-1}) \frac{\delta_i^{n-1} - \delta_i^{n-2}}{\delta_{ci}} \quad (30)$$

which was computed at the end of the previous increment, $n-1$ and

4. finally, if the number N'_p of these components is lower than or equal to $N_p^{\max}$, the corresponding indices are all included in C_n and $N_p = N'_p$, otherwise only the first $N_p^{\max}$ indices will and $N_p = N_p^{\max}$.

Notice that point 2 relates to the choice of controlling only parameters within the 'process zone', whereas point 3 relates to the choice of controlling the 'most rapidly changing' of these.

In definition of $\Delta\hat{\delta}_i^{n-1}$, multiplication by $\text{sgn}(\delta_i^{n-1})$ makes it positive if the incremental change is 'in the direction of increasing delamination' and negative if it does not and, in Figures 5(a)–(d), four examples are shown.

The term 'normalized' relates to the division by δ_{ci} , which makes the comparison between the different $\Delta\hat{\delta}_i^{n-1}$, for different indices i , more consistent. In fact, two different indices, i and j , can correspond to different material properties. This is often the case if, even at the same point, i and j relate to different modes with $\delta_{ci} \neq \delta_{cj}$. Also, this can occur if the material

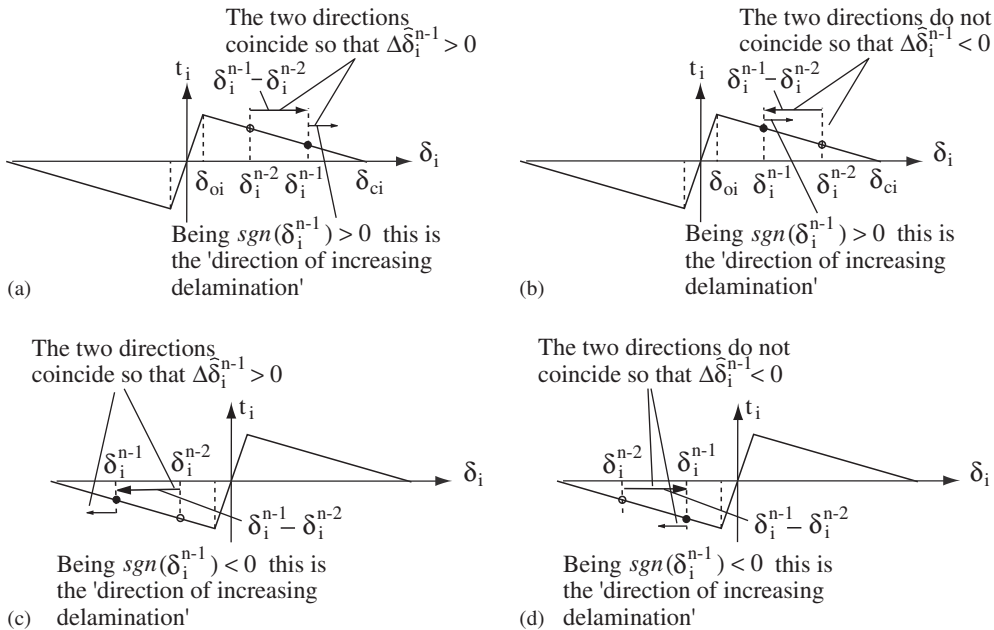


Figure 5. (a) $\text{sgn}(\delta_i^{n-1})(\delta_i^{n-1} - \delta_i^{n-2}) > 0$; (b) $\text{sgn}(\delta_i^{n-1})(\delta_i^{n-1} - \delta_i^{n-2}) < 0$; (c) $\text{sgn}(\delta_i^{n-1})(\delta_i^{n-1} - \delta_i^{n-2}) > 0$; and (d) $\text{sgn}(\delta_i^{n-1})(\delta_i^{n-1} - \delta_i^{n-2}) < 0$.

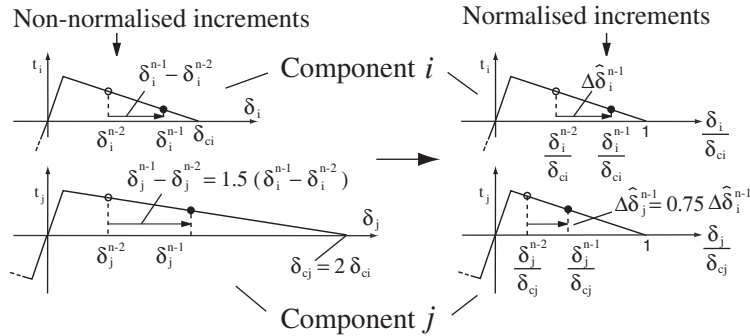


Figure 6. Example of comparison between normalized components.

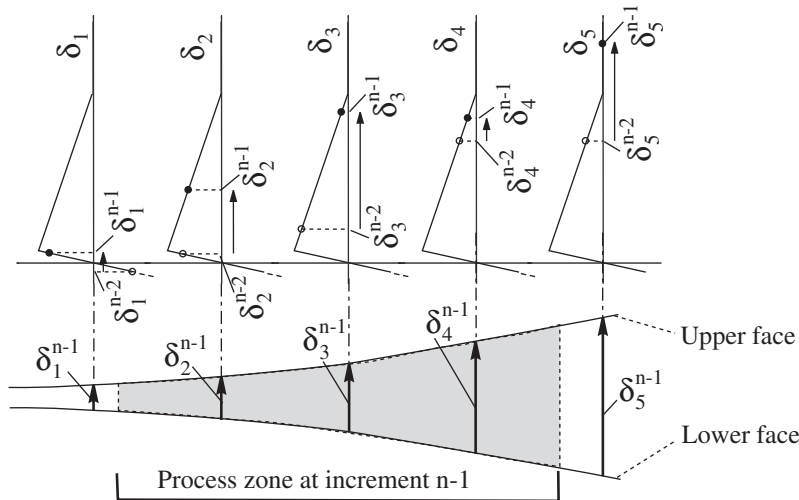


Figure 7. Local control: an example.

properties vary between different points of the same interface, or if different interfaces with different properties simultaneously delaminate. In Figure 6, for example, two components, i and j , are considered and, for simplicity, they are two cases with $\text{sgn}(\delta_i^{n-1}) = \text{sgn}(\delta_j^{n-1}) = 1$, in order to remove the sign issue. The incremental change $\delta_i^{n-1} - \delta_i^{n-2}$ is lower than $\delta_j^{n-1} - \delta_j^{n-2}$.

However, the 'normalized' value $\Delta \hat{\delta}_i^{n-1}$ turns out to be greater than $\Delta \hat{\delta}_j^{n-1}$.

A further example in Figure 7 shows how the set, C_n , is built up for a (hypothetical) case with only five components. For simplicity, mode I is considered, so that no sign issue exists, and $\delta_{ci} = 1$ for all i , so that division by δ_{ci} is immaterial. The components, δ_1 and δ_5 , are ruled out, because they are not in the process zone as $\delta_1^{n-1} < \delta_{o1}$ and $\delta_5^{n-1} > \delta_{c5}$, so that $N'_p = 3$. The remaining components are ordered following $\Delta \hat{\delta}_3 > \Delta \hat{\delta}_2 > \Delta \hat{\delta}_4$ and the set C_n is chosen as follows. If $N_p^{\max} = 1$, then $C_n = \{3\}$ and $N_p = 1$. If $N_p^{\max} = 2$, then $C_n = \{3, 2\}$ and $N_p = 2$. If $N_p^{\max} \geq 3$, then $C_n = \{3, 2, 4\}$ and $N_p = 3$.

After choosing the controlled components, that is determining C_n , the constraint equation is then written as follows:

$$\mathbf{a}_n \cdot \Delta \mathbf{p} = \sum_{i \in C_n} w_i \operatorname{sgn}(\delta_i^{n-1})(\delta_i - \delta_i^{n-1}) = \Delta I_n \quad \text{with } w_i = \frac{1}{\frac{\delta_{ci}}{\sum_{i \in C_n} \frac{1}{\delta_{ci}}}} \quad (31)$$

Using (29) the explicit expression of $\mathbf{a}_n$ is obtained:

$$\mathbf{a}_n = \sum_{i \in C_n} w_i \operatorname{sgn}(\delta_i^{n-1}) \mathbf{B}_{(\text{row } i)} \quad (32)$$

In analogy with the convenience of using the ‘normalized’ components for the construction of the control function, the weighting parameters, w_i , are introduced because the control parameters, as well as their number N_p , typically change from one increment to the next and the new parameters might correspond to different values of δ_{ci} .

In the special case that δ_{ci} is the same for all of the components of the process zone, then C_n would include the components with the largest values of $\operatorname{sgn}(\delta_i^{n-1})\Delta\delta_i^{n-1}$, and the constraint equation becomes:

$$\mathbf{a}_n \cdot \Delta \mathbf{p} = \frac{1}{N_p} \sum_{i \in C_n} \operatorname{sgn}(\delta_i^{n-1})(\delta_i - \delta_i^{n-1}) = \Delta I_n \quad (33)$$

In addition, if only one component, δ_i , is controlled ($N_p = 1$), the constraint equation simplifies to:

$$\mathbf{a}_n \cdot \Delta \mathbf{p} = \operatorname{sgn}(\delta_i^{n-1})(\delta_i - \delta_i^{n-1}) = \Delta I_n \quad (34)$$

The procedure outlined above cannot be used when there are no integration points in the process zone. This always occurs at the beginning of the analysis, when the whole interface is undamaged, and may occur at the end of the analysis, when the interface is fully damaged, or during the analysis, if the mesh is too coarse so that the size of the process zone turns out to be less than the space between two integration points. In all of these cases, one single parameter is chosen as control function. If there are fully damaged points, we choose among them the component with the minimum positive value of $\Delta\delta_i^{n-1}$. Otherwise, all of the points are still undamaged and the component with the maximum value of $\Delta\delta_i^{n-1}$ is taken.

For the first increment, not only are all of the points still undamaged, but also no previous increment exists. A dummy predictor iteration is thus introduced with a unit value of $\delta\lambda$ and the resulting relative-displacement components computed are denoted by $(\delta_i)_{\text{pred}}$. In the strategy used to determine C_n , previously explained, δ_i^{n-1} is replaced by $(\delta_i)_{\text{pred}}$ and δ_i^{n-2} is set to zero. Furthermore, in Equation (31), $\delta_i^{n-1} = 0$ as the initial displacements are null.

The local-control method was initially implemented with no line searches (NLS), and the resulting algorithm will be indicated as the ‘NLS method’.

Remark 3.2

Introducing geometric non-linearity yields a non-linear relationship between $\mathbf{p}$ and δ , instead of (29). This, in turn, results in a non-linear control function and also makes the combination with line searches more complicated. This is the reason why here we have restricted

the treatment to geometrically linear problems, although we plan to extend the procedure to cover geometric non-linearity (as is required for problems combining delamination and buckling [35]).

Remark 3.3

The procedure used to build up $\mathbf{a}_n$ is such that the control function is usually increasing on the equilibrium path in the vicinity of the initial point $\mathbf{p}_{n-1}$. However, the possibility of having limit points close to $\mathbf{p}_{n-1}$ is not mathematically excluded, although it never occurred in the numerical work.

3.2. The single line-search (SLS) method

In comparison with the cylindrical arc-length (CAL) method, proposed in Reference [24] and used in Reference [18] for the analysis of delamination in composites, the NLS method was often found to be more efficient. However, in a number of cases it was also more unstable, as is shown in Section 4. The lack of robustness is typically related to overcoming sharp limit points and, in the lights of previous work by the second author on similar problems involving non-smooth non-linearities [26], the use of line searches seemed a possible remedy.

In this section and in Section 3.3 the combination of the local-control method with line searches is studied, and we will always consider a corrector iteration.

The classical ‘predictor/corrector’ iterative method, in which the load factor is kept fixed to a given value, λ_o , within each increment, yields, at each iteration, an iterative change of the displacements, $\delta\mathbf{p}$, which is the solution to system (16) for $\delta\lambda=0$ if the full Newton method is used. Classically, a line search is performed to ‘improve’ the displacement update, in accordance with the formula, $\mathbf{p}(\eta)=\mathbf{p}_o+\eta\delta\mathbf{p}$, which defines a straight line, z , of the displacement space, along which the line search is conducted. If a total potential energy ψ exists, the evaluation of ψ over z yields the compound function defined by $\psi_z(\eta)=\psi[\mathbf{p}(\eta),\lambda_o]$. The optimal line-search parameter, $\bar{\eta}$, can therefore be computed by enforcing the stationarity of ψ_z :

$$\left. \frac{d\psi_z}{d\eta} \right|_{\bar{\eta}} = \left. \frac{\partial\psi}{\partial\mathbf{p}} \right|_{\mathbf{p}(\bar{\eta})} \cdot \left. \frac{\partial\mathbf{p}}{\partial\eta} \right|_{\bar{\eta}} = \mathbf{g}(\bar{\eta}) \cdot \delta\mathbf{p} = 0 \quad \text{with} \quad \mathbf{g}(\eta) = \mathbf{q}_i(\mathbf{p}_o + \eta\delta\mathbf{p}, \lambda_o) - \mathbf{q}_{\text{ext}}(\lambda_o) \quad (35)$$

The latter amounts to the application of the principle of stationarity of ψ to an ideal one-degree-of-freedom system, which is constrained to move along z . If ψ is locally convex, then $\mathbf{p}(\bar{\eta})$ represents a local minimum of ψ along the direction of $\delta\mathbf{p}$. Equation (35) is never solved exactly, but rather a ‘slack’ numerical tolerance is introduced [27, 29].

Such a nice theoretical framework cannot immediately be extended to a general path-following method. Firstly, if the load factor changes, an update formula of the type $\lambda(\eta)=\lambda_o+\delta\lambda(\eta)$ must also be considered, and $(\mathbf{p}(\eta),\lambda(\eta))$ would define a curve c of the $(N+1)$ -dimensional space of the unknowns. If a total potential energy ψ exists, the evaluation of ψ over c yields the compound function defined as $\psi_c(\eta)=\psi[\mathbf{p}(\eta),\lambda(\eta)]$. Minimization of ψ_c over c would no longer represent an equilibrium equation and cannot therefore be used for the evaluation of the optimal step length, $\bar{\eta}$. This issue also arises for the conventional strategy, with the load fixed to λ_o , if a total potential energy does not exist. In either case, this shortcoming can simply be by-passed by means of a weaker interpretation of Equation (35), which can be viewed as a projection of the equilibrium equation $\mathbf{g}(\eta)=\mathbf{0}$ over the

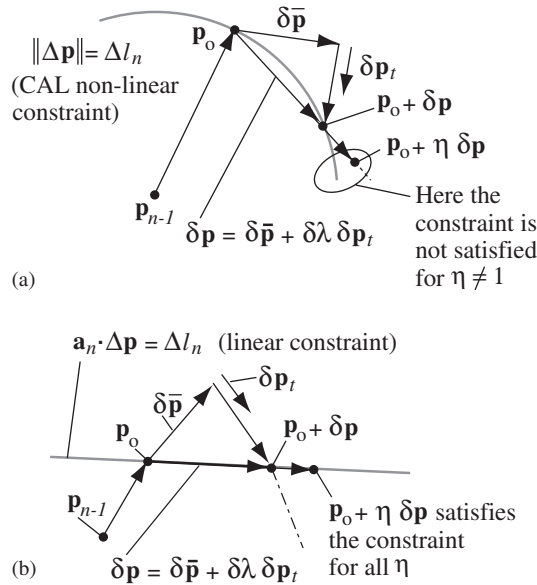


Figure 8. (a) $\mathbf{p}_o + \eta \delta \mathbf{p}$ generally does not satisfy a non-linear constraint for $\eta \neq 1$; and
(b) $\mathbf{p}_o + \eta \delta \mathbf{p}$ satisfies a linear constraint for all η .

search direction, $\delta \mathbf{p}$. In fact, equation $\mathbf{g}(\eta) \cdot \delta \mathbf{p} = 0$ can be written regardless of any stationarity principle and then also in conjunction with a path-following method.

A second problem is that, if the arc-length constraint is satisfied for $\eta = 1$, that is by $(\mathbf{p}(1), \lambda(1))$, it may not be satisfied for $\eta \neq 1$ if the constraint is non-linear, as is illustrated in Figure 8(a) in relation to the CAL method. This issue was addressed in References [25–29, 39] and we refer to this work if a general type of control function is to be considered. However, Equation (31), as well as any other linear constraint equation, represents a special and simpler case. In fact, since we are dealing with a corrector iteration, we can assume that the arc length was satisfied by the non-converged values $(\mathbf{p}_o, \lambda_o)$, in that $\mathbf{a} \cdot \Delta \mathbf{p}_o = \Delta l_n$, and the arc-length equation becomes:

$$\mathbf{a} \cdot \delta \mathbf{p} = 0 \quad (36)$$

Accordingly, if $\delta \mathbf{p}$ satisfies Equation (36), so does $\eta \delta \mathbf{p}$, regardless of the value of $\delta \lambda$, as is shown by Figure 8(b). Any choice of $\delta \lambda$ would therefore make the arc-length constraint satisfied and the general procedure outlined in References [25, 29] will be followed here, so that the following linear update formulae are used:

$$\mathbf{p}(\eta) = \mathbf{p}_o + \eta \delta \mathbf{p} \quad \lambda(\eta) = \lambda_o + \eta \delta \lambda \quad (37)$$

The combination of the local-control arc-length method described in Section 3.1 with the ‘conventional’ line search is thus as follows. The following scalar product is defined as a function of the step length, η :

$$f(\eta) = \mathbf{g}(\eta) \cdot \delta \mathbf{p} = [\mathbf{q}_i(\mathbf{p}_o + \eta \delta \mathbf{p}, \lambda_o + \eta \delta \lambda) - \mathbf{q}_{\text{ext}}(\lambda_o + \eta \delta \lambda)] \cdot \delta \mathbf{p} \quad (38)$$

and the orthogonality condition, $\mathbf{g}(\eta) \cdot \delta \mathbf{p} = 0$, is ‘relaxed’ by introducing a numerical tolerance, γ_{ls} , as follows:

$$|r(\eta)| < \gamma_{\text{ls}} \quad \text{with} \quad r(\eta) = \frac{f(\eta)}{f(0)} = \frac{\mathbf{g}(\eta) \cdot \delta \mathbf{p}}{\mathbf{g}_0 \cdot \delta \mathbf{p}} \quad (39)$$

A solution to (39) is computed by an iterative procedure which will be described in the next section. It will be referred to as a ‘ η -line search’ in order to distinguish it from an other type of line search that will be introduced later.

Remark 3.4

It is worth noting that $r(0) = 1$ and that using Equations (16)–(19) a descent direction for r in a neighborhood of $\eta = 0$ is always obtained. In fact, differentiating at $\eta = 0$, one gets:

$$\begin{aligned} f'(0) &= \left(\frac{\partial \mathbf{q}_i}{\partial \mathbf{p}} \bigg|_{(\mathbf{p}_0, \lambda_0)} \delta \mathbf{p} + \frac{\partial \mathbf{q}_i}{\partial \lambda} \bigg|_{(\mathbf{p}_0, \lambda_0)} \delta \lambda - \frac{\partial \mathbf{q}_{\text{ext}}}{\partial \lambda} \bigg|_{\lambda_0} \delta \lambda \right) \cdot \delta \mathbf{p} \\ &= (\mathbf{K}_{\text{to}} \delta \mathbf{p} - \delta \lambda \mathbf{q}'_0) \cdot \delta \mathbf{p} = -\mathbf{g}_0 \cdot \delta \mathbf{p} = -f(0) \end{aligned} \quad (40)$$

having used Equations (16) and (17). Dividing the above relationship by $f(0)$ yields $r'(0) = -1$. This is emphasized in Figure 9, where the tangent to the curve $r(\eta)$ at $\eta = 0$ is drawn.

This result shows that it would always be possible to find a positive value of the step length which decreases r to a value lower than $r(0) = 1$. However, on the grounds of our numerical experience, a simple reduction of r is not enough to say that a satisfactory step-forward towards convergence has been achieved, whereas it seems very important that, on the direction taken, it is possible to reduce r to a negative value.

3.2.1. The η -line-search. The ratio $r(\eta)$ is initially evaluated at $\eta = 1$. In many cases it is found that $|r(1)| < \gamma_{\text{ls}}$, whereby no line-search iterations are needed at all. Hence, we here assume that $|r(1)| \geq \gamma_{\text{ls}}$.

The η -line search algorithm yields a sequence of step lengths, $\{\eta_p\}$, and corresponding values $\{r_p = r(\eta_p)\}$, with $p = 0, 1, \dots$. We set $\eta_0 = 0$, so that $r_0 = r(0) = 1$. Then, we set $\eta_1 = 1$, so that $r_1 = r(1)$ is also known after what can be considered as the first η -line-search iteration. Convergence is achieved at iteration h if $|r_h| < \gamma_{\text{ls}}$. Clearly in the applications it is necessary to set a maximum number of allowed line-search iterations, N_{ls} . Accordingly, when here and in the following developments a line-search algorithm is said to ‘converge’, it will be understood that it converges ‘provided that a large enough N_{ls} is chosen’.

After the h th iteration of the search algorithm, h step lengths, η_p , and corresponding values, r_p , are known, with $p = 1, \dots, h$. If there is any negative value, r_q , with $q \leq h$, then interpolation is used, normally with a ‘regula falsi’ scheme [26, 29]: denoting by η_i and η_j the minimum and maximum step lengths corresponding to a negative and a positive r , respectively, a linear interpolation of r_i and r_j is performed (Figure 9(a)). However, because of the strong non-linearity often found in the function r , it is convenient to define a limit value, R_{lim} , of the maximum of $|r_i/r_j|$ and $|r_j/r_i|$, beyond which a ‘mid-point-like’ interpolation scheme is adopted. In the applications, the value, $R_{\text{lim}} = 5.0$, and the formula, $\eta_{h+1} = (1 - a)\eta_j + a\eta_i$, has been used, with $a = 0.3$ if $|r_j| < |r_i|$ and $a = 0.7$ in the opposite case. In Figure 9(b) an

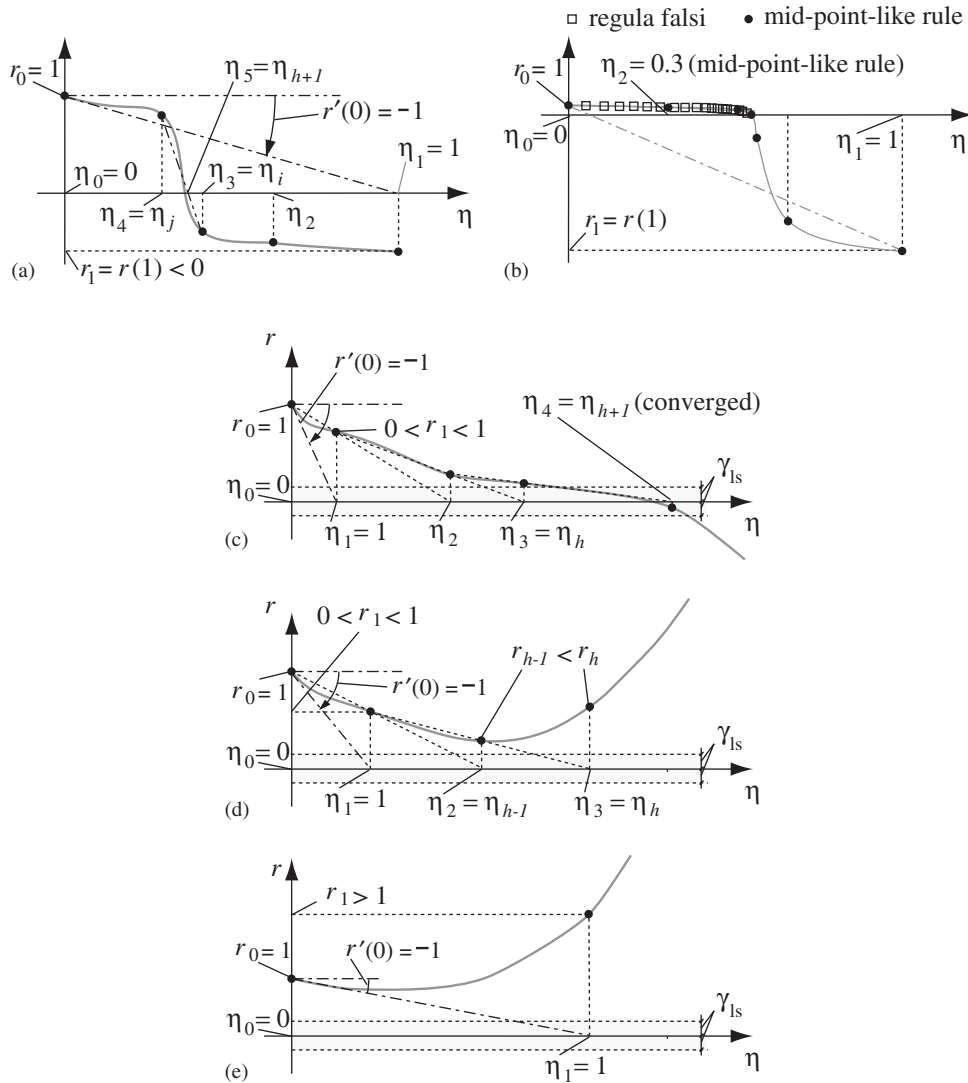


Figure 9. (a) 'Regula falsi' scheme ($h=4$); (b) mid-point-like rule; (c) $r_{h-1} > r_h > \gamma_{ls}$: extrapolate ($h=3$); (d) $\gamma_{ls} < r_{h-1} < r_h$ and convergence of the η -line search is impossible; and (e) $r_1 > 1$ and convergence of the η -line search is impossible.

example is shown, with $|r_j| < |r_i|$, where the use of the mid-point-like rule yields many less η -line-search iterations.

If all of the values, r_p , are positive, this means that extrapolation was always used in the previous $h-1$ line-search iterations, so that $\eta_0=0 < \eta_1=1 < \dots < \eta_{h-1} < \eta_h$. Linear extrapolation is then used to compute η_{h+1} if $r_h < r_{h-1}$, as in Figure 9(c). A maximum value, $\eta_{\max}$ is set, and the case when $\eta_{h+1} > \eta_{\max}$ is treated as if $r_h > r_{h-1}$. No minimum value for the step length is prescribed.

The case $r_h > r_{h-1}$ (Figure 9(d)) is more difficult to treat. In principle, this is only because simple interpolation or extrapolation formulae are not available. In practice, this often occurs when convergence of the η -line search is impossible because any reasonable step length would give $r > \gamma_{ls}$. This very often occurs after the first iteration ($r_1 > 1$), as is the case of Figure 9(e). Two strategies are considered here. The first strategy consists of abandoning the η -line search, assuming $\eta = 1$, with the hope of ‘getting out of troubled waters’ via other iterations with different directions. The implementation of this approach in conjunction with the local-control method of Section 3.1 will be referred to as the ‘single line-search’ (SLS) method. The second strategy will be fully detailed in the next section.

Remark 3.5

The η -line search algorithm used here is different from that described in References [25, 29], in particular because it allows extrapolation schemes to be used, whereas in References [25, 29] no line search is conducted if $r_1 > 0$.

3.3. The double-line-search (DLS) method

Numerical investigations, reported later, indicated that the SLS method described in the previous section is more robust than the NLS method, showing the beneficial effect of the η -line searches. Nevertheless, overcoming sharp limit points sometimes entailed a considerable number of increments at which the iterative procedure failed to converge. In those cases a reduction of the increment size was necessary and, on some occasions, the maximum number of increment cuts was reached and the analysis had to be terminated. These problems often occur when the η -line search fails to converge, as for the examples of Figures 9(d) or (e). In this section a new procedure, developed to handle such cases, is illustrated. It is indicated as a double-line-search (DLS) method because, within each iteration, it can involve two types of line-search procedure, as is summarized next and fully detailed later.

With the SLS method the iterative change of the load factor, $\delta\lambda$, is computed only once, via Equation (31), and is then kept fixed during the η -line search. Conversely, with the DLS method $\delta\lambda$ may change. Accordingly, $\delta\mathbf{p}$ also changes as it is a function of $\delta\lambda$ through Equation (18), and the scalar product (38) is a function of both η and $\delta\lambda$. In order to emphasize this aspect, a different notation will be used, whereby, instead of (38) and (39), we write:

$$\begin{aligned} h(\delta\lambda, \eta) &= \mathbf{g}(\delta\lambda, \eta) \cdot \delta\mathbf{p}(\delta\lambda) \\ &= [\mathbf{q}_i(\mathbf{p}_o + \eta\delta\bar{\mathbf{p}} + \eta\delta\lambda\delta\mathbf{p}_t, \lambda_o + \eta\delta\lambda) - \mathbf{q}_{\text{ext}}(\lambda_o + \eta\delta\lambda)] \cdot (\delta\bar{\mathbf{p}} + \delta\lambda\delta\mathbf{p}_t) \end{aligned} \quad (41)$$

The orthogonality condition, $\mathbf{g}(\delta\lambda, \eta) \cdot \delta\mathbf{p}(\delta\lambda) = 0$, is now relaxed as follows:

$$|h(\delta\lambda, \eta)| < \gamma_{ls} \quad \text{with} \quad \rho(\delta\lambda, \eta) = \frac{h(\delta\lambda, \eta)}{h(\delta\lambda, 0)} \quad (42)$$

3.3.1. Basic steps of the algorithm. Firstly, $\delta\mathbf{p}$ and $\delta\lambda$ are computed using (16)–(19) and (31). Throughout this section the value of $\delta\lambda$ computed via Equation (31) will be denoted by $\delta\lambda_a$.

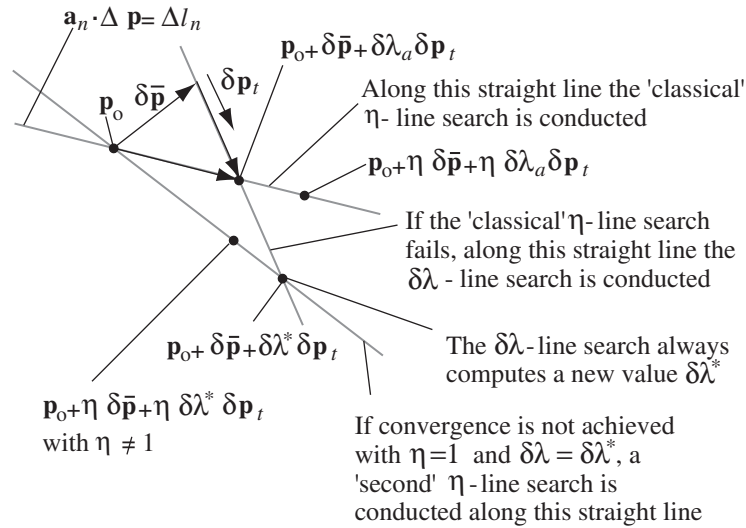


Figure 10. Scheme of the sequence of line searches entailed by the DLS method.

Then, $\rho(\delta\lambda_a, 1) = r(1)$ is evaluated. If $|\rho(\delta\lambda_a, 1)| < \gamma_{ls}$, no line search is introduced at all, the values are updated using formula (14) and the algorithm steps forward to the next iteration, as with the NLS method.

If $|\rho(\delta\lambda_a, 1)| \geq \gamma_{ls}$, the 'classical' η -line search is implemented as described in Section 3.2.1, with $\delta\lambda$ constantly equal to $\delta\lambda_a$, and η changing with a view to fulfilling inequality (42). If the η -line search converges so that (42) is satisfied, the values are updated using formula (37) and the algorithm steps forward to the next iteration, as with the SLS method.

If the η -line search fails to converge, as for the cases of Figures 9(d) or (e), we abandon it and set $\eta = 1$. However, instead of stepping forward to the next iteration with the new updated values, no update is undertaken and an alternative iterative procedure is applied. This time η is maintained at unity while $\delta\lambda$ is varied with a view to fulfilling inequality (42). With $\delta\lambda$ varying, $\delta\mathbf{p}$ draws a straight line, because of formula (18). The iterative procedure, fully detailed later, will thus be referred to as a ' $\delta\lambda$ -line search'. At the end of the $\delta\lambda$ -line search, the new value of $\delta\lambda$ will be denoted by $\delta\lambda^*$. For many cases, convergence is directly achieved, in that inequality (42) is directly satisfied with $\eta = 1$ and $\delta\lambda = \delta\lambda^*$. However, if this does not occur, it will be shown that a 'second' η -line search will be successful. This time $\delta\lambda$ is maintained as $\delta\lambda^*$, whereas η changes. In all cases, the final step length will be denoted by η^* and, eventually, inequality (42) is always satisfied by the two values, $\delta\lambda^*$ and η^* . The procedure is graphically summarized in Figure 10.

At the end of the iteration, the displacements and the load factor are updated so that:

$$\mathbf{p} = \mathbf{p}_o + \eta^*(\delta\bar{\mathbf{p}} + \delta\lambda^*\delta\mathbf{p}_t) \quad \lambda = \lambda_o + \eta^*\delta\lambda^* \quad (43)$$

and the algorithm steps forward to the next iteration. However, since the local-control constraint (31) has been violated, the increment length will 'temporarily' be changed in the successive iterations of the increment, as will be explained in Section 3.3.4.

3.3.2. *The $\delta\lambda$ -line-search.* Let us fix $\eta=1$ and use the notation $\rho^1(\delta\lambda)=\rho(\delta\lambda,1)$, so that:

$$\rho^1(\delta\lambda) = \frac{[\mathbf{q}_i(\mathbf{p}_o + \delta\tilde{\mathbf{p}} + \delta\lambda\delta\mathbf{p}_t, \lambda_o + \delta\lambda) - \mathbf{q}_{\text{ext}}(\lambda_o + \delta\lambda)] \cdot (\delta\tilde{\mathbf{p}} + \delta\lambda\delta\mathbf{p}_t)}{\mathbf{g}_o \cdot (\delta\tilde{\mathbf{p}} + \delta\lambda\delta\mathbf{p}_t)} \quad (44)$$

With a ‘ $\delta\lambda$ -line search’ we compute a sequence of corresponding values $\{\delta\lambda_p\}$ and $\{\rho_p^1 = \rho^1(\delta\lambda_p)\}$, with $p=1,2,\dots$.

The denominator of (44) is null at the following value, $\delta\tilde{\lambda}$:

$$\delta\tilde{\lambda} = -\frac{\mathbf{g}_o \cdot \delta\tilde{\mathbf{p}}}{\mathbf{g}_o \cdot \delta\mathbf{p}_t} \quad (45)$$

and its sign for $\delta\lambda > \delta\tilde{\lambda}$ is the same as that of the product:

$$h_{\text{ot}} = \mathbf{g}_o \cdot \delta\mathbf{p}_t \quad (46)$$

We rule out the case that the nominator of (44) is also null at $\delta\tilde{\lambda}$. Hence, since both the numerator and the denominator of (44) are continuous functions of $\delta\lambda$, $\delta\tilde{\lambda}$ is the only point of discontinuity for ρ^1 . The function ρ^1 has a vertical asymptote at $\delta\tilde{\lambda}$, which is associated with the iterative change, $\delta\tilde{\mathbf{p}} = \delta\tilde{\mathbf{p}} + \delta\tilde{\lambda}\delta\mathbf{p}_t$. In order to investigate the behaviour of ρ^1 in the neighbourhood of $\delta\tilde{\lambda}$, the nominator of (44) at $\delta\tilde{\lambda}$ is then required:

$$\tilde{h} = [\mathbf{q}_i(\mathbf{p}_o + \delta\tilde{\mathbf{p}}, \lambda_o + \delta\tilde{\lambda}) - \mathbf{q}_{\text{ext}}(\lambda_o + \delta\tilde{\lambda})] \cdot \delta\tilde{\mathbf{p}} \quad (47)$$

This entails a preliminary post-processing of the solution, which is computationally equivalent to one line-search iteration. The leftmost and rightmost limits of ρ^1 at $\delta\tilde{\lambda}$ are then as follows:

$$\begin{aligned} \lim_{\delta\lambda \rightarrow \delta\tilde{\lambda}^-} \rho^1(\delta\lambda) &= \begin{cases} -\infty & \text{if } \tilde{h} \cdot h_{\text{ot}} > 0 \\ +\infty & \text{if } \tilde{h} \cdot h_{\text{ot}} < 0 \end{cases} \\ \lim_{\delta\lambda \rightarrow \delta\tilde{\lambda}^+} \rho^1(\delta\lambda) &= \begin{cases} +\infty & \text{if } \tilde{h} \cdot h_{\text{ot}} > 0 \\ -\infty & \text{if } \tilde{h} \cdot h_{\text{ot}} < 0 \end{cases} \end{aligned} \quad (48)$$

Recalling that we denote by $\delta\lambda_a$ the solution to Equation (31), let us assume, without loss of generality, that $\delta\lambda_a > \delta\tilde{\lambda}$. It is then easy to verify that the case $\delta\lambda_a < \delta\tilde{\lambda}$ can be treated in a perfectly analogous way. Furthermore, it is clear that $\rho^1(\delta\lambda_a) > 0$, since otherwise the $\delta\lambda$ -line search would not have been started.

Depending on the value of $\tilde{h} \cdot h_{\text{ot}}$, two cases occur: ‘case a’ if $\tilde{h} \cdot h_{\text{ot}} < 0$ and ‘case b’ if $\tilde{h} \cdot h_{\text{ot}} > 0$. Figure 11(a) relates to case a: continuity of ρ^1 in $(\delta\tilde{\lambda}, +\infty)$ ensures that there exists at least one value $\delta\lambda \in (\delta\tilde{\lambda}, \delta\lambda_a)$ whereby $\rho^1(\delta\lambda) = 0$. Accordingly, $\delta\lambda_a$ is taken as the first value of the $\delta\lambda$ -line search, so we set $\delta\lambda_1 = \delta\lambda_a$. We then proceed by computing a sequence

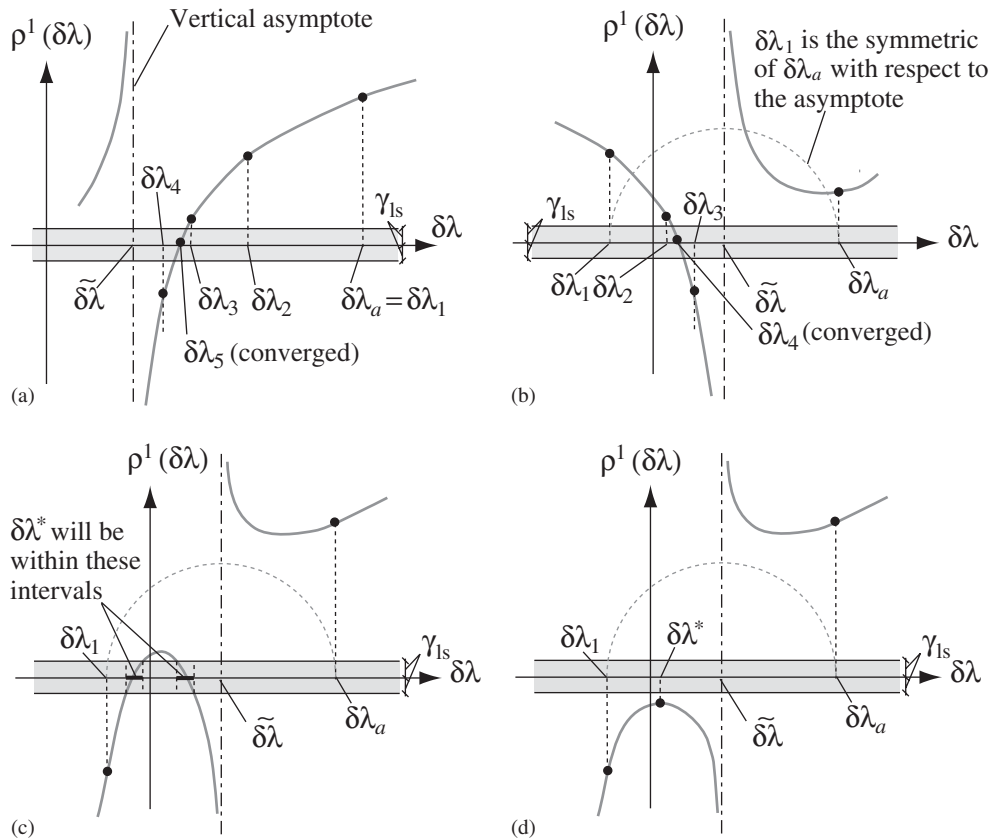


Figure 11. (a) Case a: $\tilde{h} \cdot h_{ot} < 0$; (b) case b1: $\tilde{h} \cdot h_{ot} > 0$ and $\rho_1^1 > 0$; (c) case b2: $\tilde{h} \cdot h_{ot} > 0$ and $\rho_1^1 < 0$ but then convergence is achieved; and (d) case b3: $\tilde{h} \cdot h_{ot} > 0$ and a negative local maximum for ρ^1 is found at $\delta\lambda^*$.

of values $\delta\lambda_{h+1} \in (\tilde{\delta\lambda}, \delta\lambda_h)$, with a simple mid-point scheme:

$$\delta\lambda_{h+1} = \frac{\tilde{\delta\lambda} + \delta\lambda_h}{2} \quad (49)$$

until $\rho_h^1 < 0$. Then we can continue the $\delta\lambda$ -line search using interpolation, and convergence will be eventually achieved. The interpolation scheme is analogous to that used for the η -line search: denoting by $\delta\lambda_i$ and $\delta\lambda_j$ the minimum and maximum values corresponding to a positive and a negative ρ^1 , respectively, a linear interpolation of ρ_i^1 and ρ_j^1 is normally applied. Again, a limit value for the maximum of $|\rho_i^1/\rho_j^1|$ and $|\rho_j^1/\rho_i^1|$ is defined, beyond which a 'mid-point-like' interpolation scheme is adopted. In the applications, the limit value was taken equal to 3.0 and the formula, $\eta_{h+1} = (1 - a)\eta_j + a\eta_i$, has been used, with $a = 0.3$ if $|\rho_j| < |\rho_i|$ and $a = 0.7$ in the opposite case.

In the example of Figure 11(a), formula (49) was used to evaluate $\delta\lambda_2$, $\delta\lambda_3$ and $\delta\lambda_4$. Then, with $\rho_4^1 < 0$ and $|\rho_4^1/\rho_3^1| < 3.0$ as well as $|\rho_3^1/\rho_4^1| < 3.0$, interpolation with the 'regula falsi' is

used to compute $\delta\lambda_5$. Since $|\rho_5^1| = |\rho(\delta\lambda_5, 1)| < \gamma_{ls}$, inequality (42) is satisfied, so that $\delta\lambda^* = \delta\lambda_5$ and $\eta^* = 1$.

Remark 3.6

Referring, for simplicity, to Figure 11(a), it is worth noting that we could stop the $\delta\lambda$ -line search at $\delta\lambda_4$, because $\rho_4^1 < 0$. This is because a new η -line search, with η changing and $\delta\lambda = \delta\lambda_4$, would converge as it would be implemented with an interpolation scheme. However, we found it more convenient, whenever possible, to continue the $\delta\lambda$ -line search until condition (42) is fulfilled.

Figures 11(b)–(d) relate to case b, for which three alternative subcases are possible, cases b1, b2 and b3. In all of the cases there is no guarantee that the function ρ^1 assumes negative values for $\delta\lambda > \tilde{\delta\lambda}$. In fact, this is unlikely. We then choose the value, $\delta\lambda_1 = 2\tilde{\delta\lambda} - \delta\lambda_a = \tilde{\delta\lambda} - (\delta\lambda_a - \tilde{\delta\lambda})$, that is the value symmetric to $\delta\lambda_a$ with respect to $\tilde{\delta\lambda}$, as a first prediction for the $\delta\lambda$ -line search. Case b1 (Figure 11(b)) occurs when a positive value, $\rho^1(\delta\lambda_1)$, is found. This case is equivalent to the previously discussed case a, except for the trivial difference that now $\delta\lambda_1 < \tilde{\delta\lambda}$, and then the same procedure as previously explained can be used to obtain convergence. If $\rho^1(\delta\lambda_1) < 0$, as for Figures 11(c)–(d), we do not know in advance whether there are values, $\delta\lambda < \tilde{\delta\lambda}$, such that $|\rho^1(\delta\lambda)| < \gamma_{ls}$. Fortunately, if $\rho^1(\delta\lambda_1) < 0$ satisfaction of (42) is desirable but not essential and, in principle, we could stop the $\delta\lambda$ -line search. This is because a new η -line search, with $\delta\lambda^* = \delta\lambda_1$, would eventually converge, as explained in Remark 3.6. However, we try to improve the choice of $\delta\lambda^*$ with a procedure detailed in Appendix A. Case b2 occurs when convergence is achieved, and b3 is the opposite case, whereby convergence is not obtained even if we cannot exclude the existence of some $\delta\lambda$ such that $|\rho^1(\delta\lambda)| < \gamma_{ls}$. For case b2 the position of the converged value, $\delta\lambda^*$, depends on the position of the initial guess $\delta\lambda_1$ (Figure 11(c)). For case b3, $\delta\lambda^*$ is typically a good approximation of a (negative) local maximum for ρ^1 . At least, this was always obtained in the numerical applications, although it is not possible to exclude other possibilities. In any case, for case b3 a value $\delta\lambda^*$ is computed such that $\rho^1(\delta\lambda^*) < 0$. Hence, only in case b3 is a new η -line search activated, with $\delta\lambda = \delta\lambda^*$, which converges to a step length, $\eta^* < 1$. In all of the other cases, $\eta^* = 1$.

3.3.3. Convergence of the double-line-search procedure. The sequence of line-search procedures entailed by the DLS method can be considered as a unique iterative algorithm and, since the tolerance γ_{ls} can be any positive value, the following convergence result holds true:

Proposition 1

Denote by $\{\delta\lambda_k, \eta_k\}$ the sequence of values generated in one iteration, with the DLS method, by the sequence of η - and $\delta\lambda$ -line searches. Provided that $\tilde{h} \neq 0$, with $\tilde{h}$ given by relationship (47), it follows that:

$$\lim_{k \rightarrow \infty} \rho(\delta\lambda_k, \eta_k) = 0$$

3.3.4. Changing the increment length. We have so far assumed that the increment length, Δl_n , is kept fixed within an increment, n , and changes from increment to increment in accordance with Equation (21). For the DLS method we comply with this rule in all of those

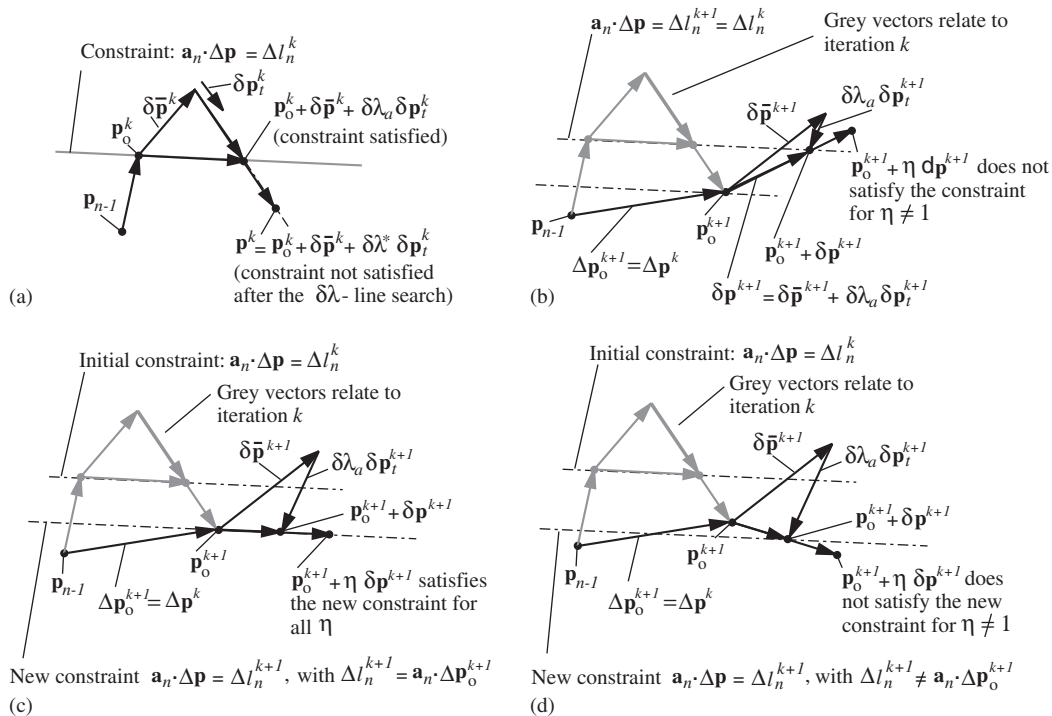


Figure 12. (a) Iteration k ; (b) iteration $k+1$, with $\Delta l_n^{k+1} = \Delta l_n^k$; (c) iteration $k+1$, with $\Delta l_n^{k+1} = \mathbf{a}_n \cdot \Delta \mathbf{p}_0^{k+1}$; and (d) iteration $k+1$, with $\Delta l_n^{k+1} \neq \mathbf{a}_n \cdot \Delta \mathbf{p}_0^{k+1}$ and $\Delta l_n^{k+1} \neq \Delta l_n^k$.

increments which do not require the use of $\delta \lambda$ -line searches. However, when the $\delta \lambda$ -line searches are called upon within increment n , we prefer to temporarily modify the increment length as is hereafter detailed. Accordingly, in this section we will denote by Δl_n the increment length initially evaluated via (21) and by Δl_n^k the value used at iteration k , which might be different from Δl_n . In addition, more than one single iteration will be considered and a superscript k will be used for all of the quantities which relate to the k th iteration.

When the $\delta \lambda$ -line search is used within iteration k , the arc-length constraint (31) is violated by the displacements, $\mathbf{p}^k$, at the end of the iteration (Figure 12(a)). Firstly, let us assume that (15) is not satisfied so that more iterations are needed and we set $\mathbf{p}_0^{k+1} = \mathbf{p}^k$ and $\lambda_o^{k+1} = \lambda^k$. We will not choose $\Delta l_n^{k+1} = \Delta l_n^k$, which would make the use of the η -line search impossible without violating Equation (31) because $\mathbf{a}_n \cdot \Delta \mathbf{p}_0^{k+1} \neq \Delta l_n^{k+1}$ (Figure 12(b)). Instead, a different value of Δl_n^{k+1} will be assumed. The preferred strategy is to set $\Delta l_n^{k+1} = \mathbf{a}_n \cdot \Delta \mathbf{p}^k = \mathbf{a}_n \cdot \Delta \mathbf{p}_0^{k+1}$. In this way, the η -line search can normally be applied within iteration $k+1$ without violating Equation (31) (Figure 12(c)), and no modification of the DLS method in the new iteration is required. An alternative option is to use a value, Δl_n^{k+1} , which is different both from Δl_n^k and from $\mathbf{a}_n \cdot \Delta \mathbf{p}^k$.

We will set $\Delta l_n^{k+1} = \mathbf{a}_n \cdot \Delta \mathbf{p}^k$ if either $\mathbf{a}_n \cdot \Delta \mathbf{p}^k > 0$ or, if $\mathbf{a}_n \cdot \Delta \mathbf{p}^k \leq 0$, at the end of iteration k an overall 'increase of delamination' is found. To this end, the total amount of damage in the structure, $D = \int_{\Omega} d \, d\Omega$, is computed, through a numerical integration of the damage

variables d (see Equation (5)) computed at all of the integration points. If a positive incremental change, ΔD_n^k , of D is found at the end of the iteration k , then we set $\Delta l_n^{k+1} = \mathbf{a}_n \cdot \Delta \mathbf{p}^k$ even if $\mathbf{a}_n \cdot \Delta \mathbf{p}^k \leq 0$. In this respect, it is worth recalling the example of Figure 3, where a negative value of $\mathbf{a}_n \cdot \Delta \mathbf{p}^k$ corresponds to a step forward in the equilibrium path. On the other hand, as observed in Remark 3.3, the local-control method does not mathematically exclude the possibility of limit points for the control function in the vicinity of the previous converged point.

If $\mathbf{a}_n \cdot \Delta \mathbf{p}^k \leq 0$ and ΔD_n^k is (numerically) null, then we set $\Delta l_n^{k+1} = 0.5^m \Delta l_n$, where m is the number of times that Δl_n has been scaled during the increment, so that the first time is $m=1$, the second time is $m=2$ and so on. In this way, a positive value of Δl_n^{k+1} is ensured and its reduced value hopefully makes convergence easier. In this last case, with $\mathbf{a}_n \cdot \Delta \mathbf{p}_0^{k+1} \neq \Delta l_n^{k+1}$, no line searches are allowed in iteration $k+1$, for the reasons discussed above and graphically illustrated in Figure 12(d). In a subsequent iteration h , even if no $\delta\lambda$ -line searches are activated, the check $\Delta D_n^h > 0$ is always performed. If $\Delta D_n^h = 0$, we use again $\Delta l_n^{h+1} = 0.5^m \Delta l_n$ and do not implement any line-search procedure within iteration $h+1$.

If, at the end of an iteration which has involved a $\delta\lambda$ -line search, the residual vector meets the convergence criterion (15), we are back onto the equilibrium path. We then consider the increment converged either if $\mathbf{a}_n \cdot \Delta \mathbf{p}^k > 0$ or, in the case that $\mathbf{a}_n \cdot \Delta \mathbf{p}^k \leq 0$, if a positive ΔD_n^k is found.

In the applications, use of negative increment lengths were not very rare, but at convergence the final increment length was always positive. This means that the possibility of limit points for the control function in the vicinity of the previous converged point is indeed very unlikely.

The modification of the increment length due to the use of $\delta\lambda$ -line searches is temporary because, after convergence, Δl_n is used in Equation (21) to compute Δl_{n+1} . This is in accordance with the observation that convergence difficulties are normally related to localized steep turns in the equilibrium path. Therefore, after overcoming such points typically in one or two increments, there is no need to keep using the modified value of Δl .

The procedure used to determine the increment length within increment n can be summarized in the following scheme (which makes use of some Fortran-like notation).

```

First iteration:  $\Delta l_n^k = \Delta l_n$ ;  $m = 1$ .
Iteration  $k + 1$ :
IF no  $\delta\lambda$ -line searches were used in the previous  $k$  iterations THEN
     $\Delta l_n^{k+1} = \Delta l_n$ 
ELSEIF a  $\delta\lambda$ -line search was used in the  $k$ th iteration THEN
    IF  $\mathbf{a}_n \cdot \Delta \mathbf{p}^k > 0$  OR  $\Delta D_n^k > 0$  THEN
         $\Delta l_n^{k+1} = \mathbf{a}_n \cdot \Delta \mathbf{p}^k$ 
    ELSE
         $\Delta l_n^{k+1} = 0.5^m \Delta l_n$ 
         $m = m + 1$ 
        no line searches within this iteration
    ENDIF
ELSE (no  $\delta\lambda$ -line search in the  $k$ th iteration, but some in the previous  $k - 1$  iterations)
    IF  $\Delta D_n^k > 0$  THEN
         $\Delta l_n^{k+1} = \Delta l_n^k$ 
    ELSE

```

```

 $\Delta l_n^{k+1} = 0.5^m \Delta l_n$ 
 $m = m + 1$ 
no line searches within this iteration
ENDIF
ENDIF

```

4. NUMERICAL RESULTS

The interface element model of Section 2 and the solution strategies of Section 3 have been implemented in a research version of the finite element code LUSAS [40]. In this section we present some numerical results for three problems of delamination of laminated composites. For all of the cases comparisons are also made with available experimental data. Depending on some integer flags in the input data, the cylindrical arc length (CAL) as described in Reference [24], the no-line-search (NLS), the single-line-search (SLS) and the double-line-search (DLS) methods have been used and compared. The last three methods all involve ‘local control’.

As part of this comparison the CPU time used on a 450 MHz Pentium PC is also given. In this respect the SLS and DLS methods have to pay a substantial cost because of the large number of line-search iterations required. This might seem a major drawback of these two methods, but it is worth noting that a big reduction of the CPU time required by the line searches would be achieved if some sort of parallel methods were used. This does not necessarily imply the use of multiple processors, but rather would simply involve an implementation of the line-search algorithms which allows different values to be tried within one single line-search iteration. Furthermore, the proposed $\delta\lambda$ -line search algorithm can also be optimized, especially for the part which involves extrapolation, for which higher order schemes might be very useful. We believe that the big improvements of the numerical performances often obtained with the SLS and the DLS methods and reported in this section make this line of research particularly interesting, and future work should be undertaken on this subject.

4.1. Analysis of a double cantilever beam specimen

A double cantilever beam (DCB) has been analysed under displacement control and with a two-dimensional model. The geometry, described in Figure 13, and the material data, given in Table I, correspond to the unidirectional 24-ply carbon-fiber laminate used by Robinson and Song in their experimental analysis reported in Reference [41]. The opening displacement, $\Delta = u_t - u_b$, is prescribed and the analysis is considered successfully terminated at increment n if $\Delta_n > 15.0$ mm.

The mesh adopted is described in Figure 14: in the four rectangle areas, regular meshes of 8-noded plane-strain elements (QPSN8) were adopted. On the initially undamaged interface, 6-noded interface elements INT6 have been used, whereas on the initially delaminated part the contact-type 6-noded interface elements INT6C, described in Section 2.1, were adopted. Although the opposite sides of the initial crack were never expected to get in contact at any equilibrated configuration of the DCB during the delamination process, this is not the case for the non-equilibrated configurations during the iterative process. The contact interface elements are then useful to avoid the algorithm converging to unrealistic solutions involving penetration.

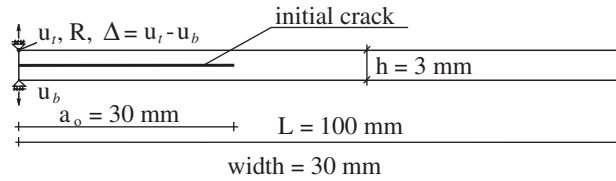


Figure 13. DCB analysis: geometry and loading.

Table I. DCB analysis: material properties [41].

Composite XAS-913C			
E_{11} (GPa)	$E_{22} = E_{33}$ (GPa)	G_{12} (GPa)	$\nu_{12} = \nu_{13} = \nu_{23}$
126	7.5	4.981	0.263
Interface			
Case	G_{c1} (N mm ⁻¹)	s_{o1} (mm)	t_{o1} (MPa)
A	0.281	5.0e - 5	28.5
B	0.281	1.0e - 4	57.0

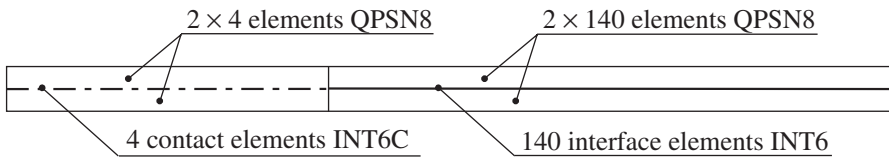


Figure 14. DCB analysis: mesh adopted.

The problem involves only mode-I delamination, so that only the material properties for mode I are needed. Two material models are considered, whose properties are detailed in Table I and correspond to those given in Reference [41] except for the tensile strength, t_{o1} and the ‘cracking’ displacement, s_{o1} .

For the numerical simulation the following solution parameters have been adopted: tolerance for the Newton iterations (see Equation (15)), $\gamma = 1.0e - 7$; maximum number of Newton iterations, $N_{it}^{max} = 15$; maximum number of increments, $N_{inc}^{max} = 200$ (never reached); desired number of iterations, $I_d = 5$; scaling factor for an increment cut, $\beta_{sc} = 0.5$; maximum allowed number of consecutive increment cuts, $N_{str}^{max} = 25$; amplifying factor for the last attempt after N_{str}^{max} increment cuts, $\beta_{amp} = 2.0$; maximum allowed change of the magnitude of the load factor at the predictor, $\Delta\lambda_{max} = 4.0$. For the SLS and DLS methods a maximum number of η -line searches or of $\delta\lambda$ -line searches, $N_{ls} = 20$, was input; the tolerance factor for the line searches was $\gamma_{ls} = 0.4$; the maximum value for the step length was set as $\eta_{max} = 5.0$. For all of the local-control methods, one relative-displacement component was always controlled, so that $N_p^{max} = 1$.

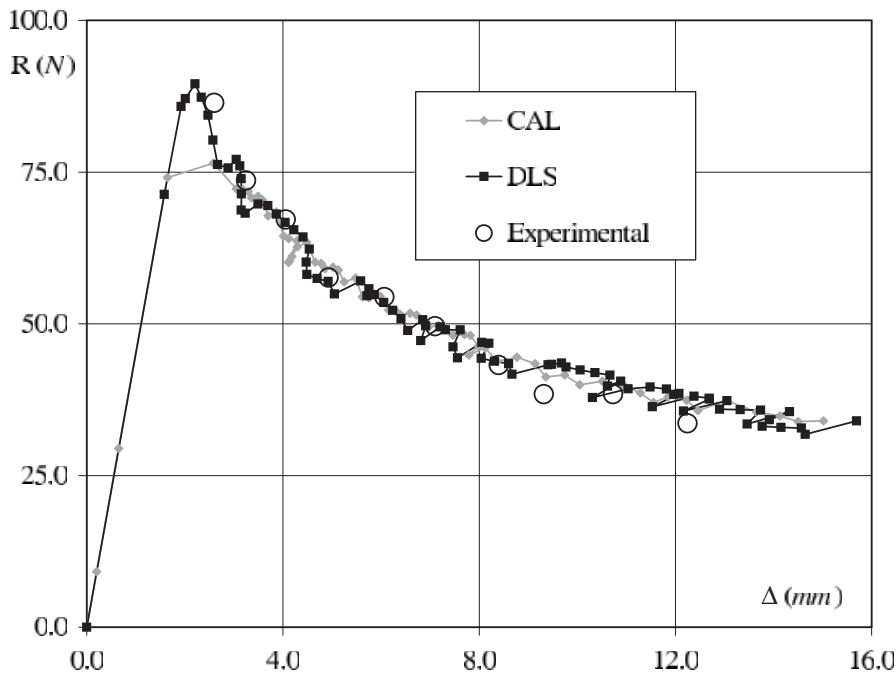


Figure 15. DCB analysis: R - Δ curves obtained with material model B and comparison with the experimental results reported in Reference [41].

The higher strength considered (material model B) is close to the tensile strength of the matrix as estimated by the experimenters. Figure 15 shows the ‘reaction/opening displacement’ ($R - \Delta$) curves obtained with the CAL and the DLS methods, which are in good agreement with the experimental results reported in Reference [41]. The other curves obtained with the NLS and the SLS methods are not plotted, as they are quite similar to that obtained with the DLS method. The lack of smoothness is due to the insufficiently refined modeling of the kinematics of the process zone, which is approximately 0.35 mm long and contains either 1 or 2 softening integration points. The superiority of the local-control schemes is testified by Table II, where the different numerical performances of the four considered algorithms are compared. Despite the decrease of the number of increment cuts from 2 to 0 obtained with the use of line searches, this is a case where the efficiency of the NLS method is satisfactory and its related CPU time is slightly lower than that of the SLS and DLS methods. Furthermore, in this case the extra complexity of the DLS method seems to be unnecessary.

By halving the tensile strength, the length of the process zone increases to approximately 1.2 mm. Accordingly, 5 softening nodal points are normally present and a much smoother curve is obtained, as shown by Figure 16. Table III confirms the superiority of the local-control schemes, which now give almost equivalent results. Line searches are then not necessary at all. It is worth noting that the length of 1.2 mm exceeds the typically observed length of the process zone and therefore the correctness of the model becomes questionable in such a case. In particular, the strain and stress fields around the crack tip are badly approximated. Nevertheless, Figure 16 also shows that the reduced tensile strength is still high enough to

Table II. DCB analysis: numerical performance with material model B.

	No. of incr.	Tot no. of iter.	No. of incr. cuts	Max. no. of consecutive incr. cuts	No. of η -line searches	No. of $\delta\lambda$ -line searches	Tot. no. of line searches	CPU time (s)	No. incr. with $\delta\lambda$ -line searches	Final Δ (mm)
CAL	63	447	23	4	—	—	—	1176	—	15.0
NLS	60	344	2	1	—	—	—	698	—	15.0
SLS	80	354	0	0	54	—	54	752	—	15.1
DLS	81	358	0	0	48	9	57	760	1	15.7

Table III. DCB analysis: numerical performance with material model A.

	No. of incr.	Tot no. of iter.	No. of incr. cuts	Max. no. of consecutive incr. cuts	No. of η -line searches	No. of $\delta\lambda$ -line searches	Tot. no. of line searches	CPU time (s)	No. incr. with $\delta\lambda$ -line searches	Final Δ (mm)
CAL	90	789	44	2	—	—	—	2105	—	15.2
NLS	50	285	0	0	—	—	—	578	—	15.2
SLS	50	285	0	0	1	—	1	578	—	15.2
DLS	50	287	0	0	3	9	12	589	1	15.2

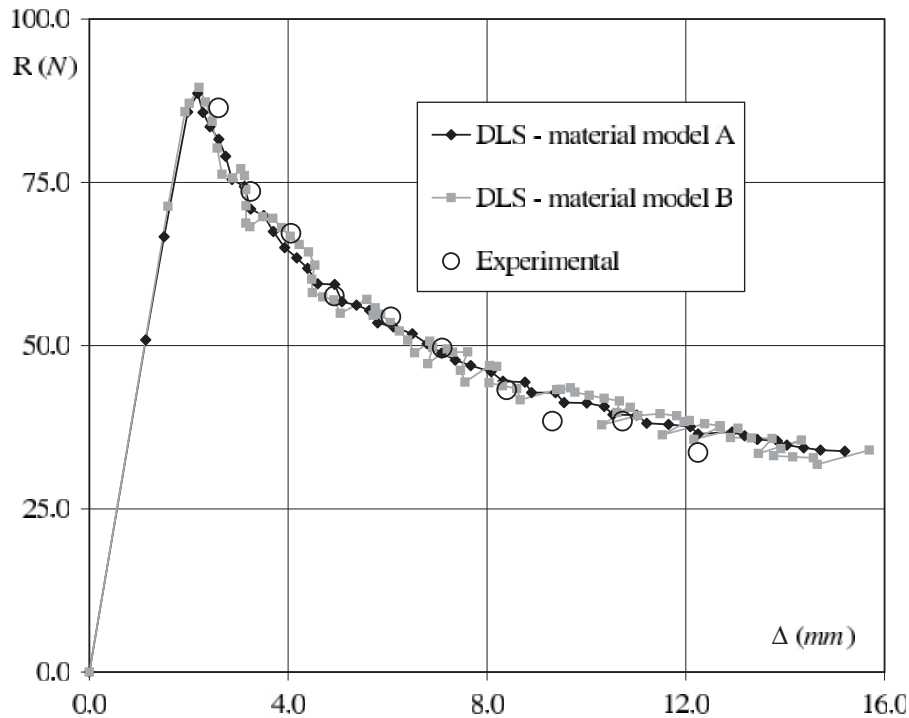


Figure 16. DCB analysis: comparison between the R – Δ curves obtained with material models A and B.

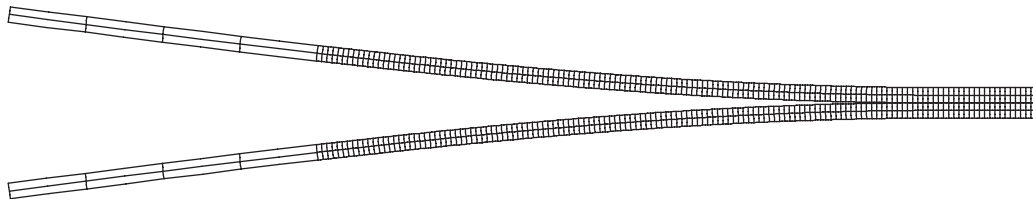


Figure 17. DCB analysis: final deformed mesh.

predict the correct value of the peak of the reaction R in the curve. Earlier work [18] showed that a further reduction of t_{01} below values of 5–10 MPa would give incorrect results also for the load-displacement curve. The deformed mesh at the end of the analysis is illustrated in Figure 17.

4.2. Multiple delamination of a composite specimen

The multiple mixed-mode delamination of a carbon-fiber laminate was analysed with a two-dimensional model under displacement control. The geometry and loading of the specimen are illustrated in Figure 18 and correspond to those considered in Reference [42]. The initial crack on the left-hand of the specimen is placed along the mid plane and will be indicated as the ‘first initial crack’. In this respect the problem seems similar to a DCB analysis, but

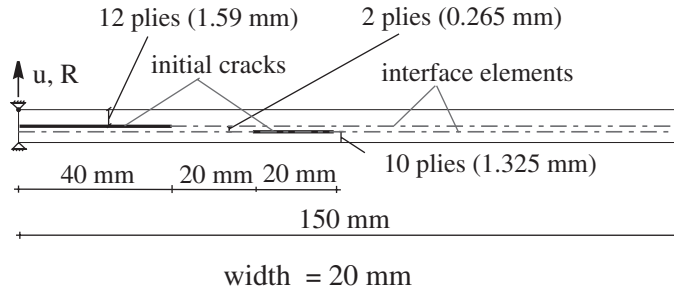


Figure 18. Multiple delamination analysis: geometry and loading.

Table IV. Multiple delamination analysis: material properties [42].

Composite Hexcel HTA913						
E_{11} (GPa)	$E_{22}=E_{33}$ (GPa)	G_{12} (GPa)		$\nu_{12} = \nu_{13} = \nu_{23}$		
115.0	8.5	4.5		0.29		
Interface						
Case	G_{c1} (N mm ⁻¹)	G_{c2} (N mm ⁻¹)	1- μ	α	t_{01} (MPa)	t_{02} (MPa)
A	0.33	0.8	5.0e-5	2	3.3	8.0
B	0.33	0.8	1.25e-3	2	16.5	40.0
C	0.33	0.8	5.0e-3	2	33.0	80.0

the presence of a ‘second initial crack’ substantially changes the response. The second initial crack is positioned 20 mm on the right of and two plies below the first crack.

Three sets of material properties, A, B and C, were considered and are illustrated in Table IV. They differ only for the values of the strengths, t_{01} and t_{02} , and for the parameter introduced in Equation (2), μ , which was changed in order to keep the penalty stiffness constant. The other material properties are those considered in Reference [42] and were experimentally measured.

Two different meshes were used and will be referred to as ‘coarse’ and ‘fine’. They are illustrated in Figure 19, where the domain is split into 15 rectangular zones in which regular

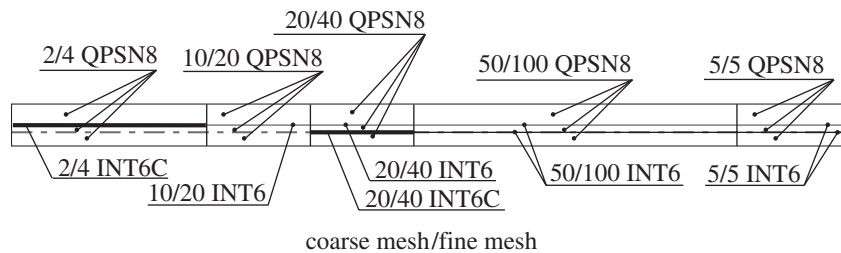


Figure 19. Multiple delamination analysis: meshes adopted.

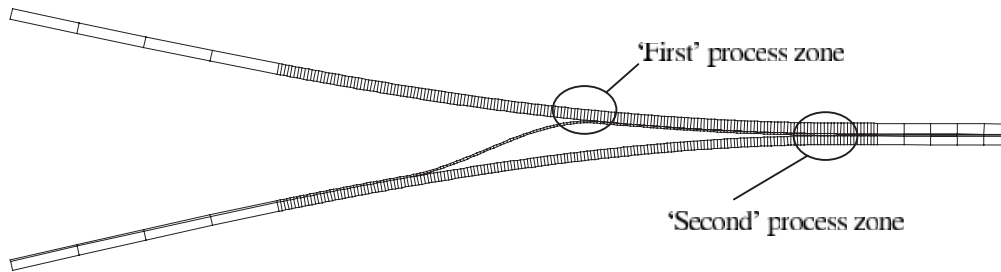


Figure 20. Multiple delamination analysis: final deformed mesh.

meshes of 8-noded plain-strain (QPSN8) elements were adopted. On all of the interfaces which were predicted to delaminate, 6-noded interface elements INT6 were positioned, whereas 6-noded contact interface elements INT6C were used to avoid penetration between the opposite sides of the initially delaminated parts.

The vertical displacement, u , of the upper-left-hand corner of the specimen was prescribed during the analysis and the corresponding reaction, R , was computed. The analysis is considered successfully terminated at increment n if $u_n > 36.0$ mm. The final deformed 'fine' mesh is shown in Figure 20.

In Figure 21 the R - u curves obtained with material models A, B and C and the DLS method are plotted and compared with the experimental results reported in Reference [42]. The analysis for the material model A was performed with the 'coarse' mesh whereas the 'fine' mesh was used for models B and C. The local load maxima were underestimated with material model A, whereas a better agreement is achieved using models B and C.

In order to describe the delamination process, reference is now made to the gray curve of Figure 21, obtained with model C. The first 'cracking point' is detected immediately on the right-hand-side of the first initial crack and at point (a) in Figure 21. However, it is only when the peak value of the reaction is reached (point (b)) that this point reaches complete delamination. Then the process zone, which between points (b) and (c) is approximately 1 mm long and will be later indicated as the 'first process zone', propagates along the upper interface following the typical behaviour of a DCB specimen. When, around point (c), the process zone reaches a distance of about 1 mm from the point vertically above the left-hand tip of the second initial crack, a sudden drop of the reaction occurs, with a 'snap-back' which is followed by an unstable path. This is in good agreement with the experimental finding of a 'dynamic jump' of the R - u curve between points lying very close to points (c) and (f). Between points (d) and (e) the first process zone reduces in size from 1.0 mm to a length of about 0.35 mm, which will remain approximately constant for the rest of the analysis. This is attributable to the reduction of the stiffness of the laminate below the first process zone as it moves over the second initial crack. In correspondence to point (f), the first 'cracking' point is detected immediately on the right-hand side of the second initial crack. Between points (f) and (g) the first process zone continues to propagate, while a 'second process zone' develops on the right-hand side of the second initial crack. At point (g), the point on the right-hand side of the second initial crack is fully delaminated so that the 'second process zone' is completely developed, with a length of about 1.0 mm which will remain approximately constant for the

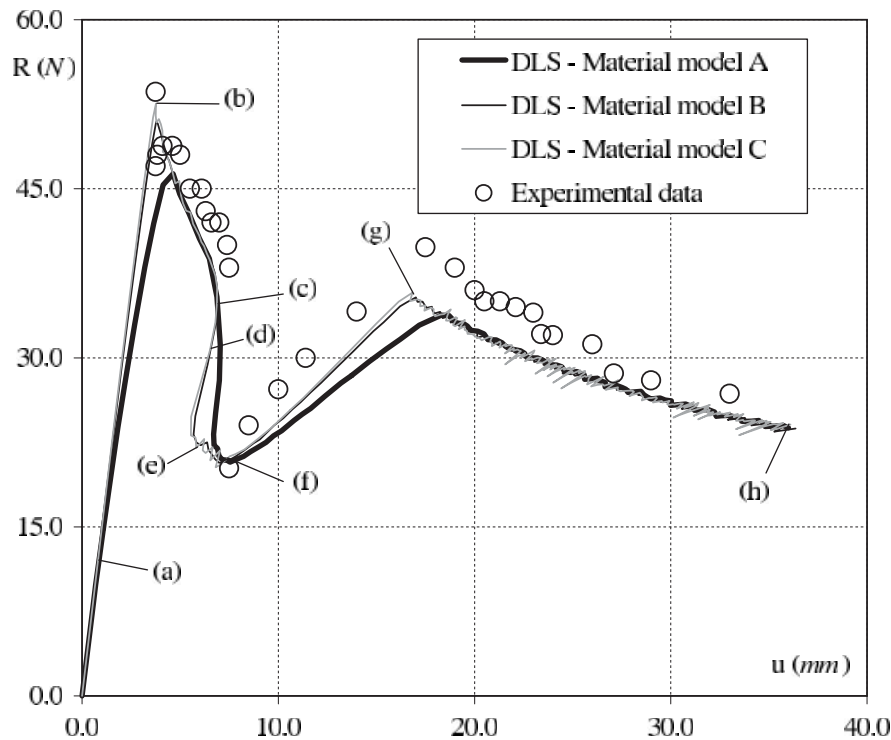


Figure 21. Multiple delamination analysis: R - u curves obtained with material models A, B, and C and comparison with the experimental results reported in Reference [42].

rest of the analysis. Between points (g) and (h) the two zones simultaneously propagate, the second zone moving between approximately 2.0 and 2.7 times faster than the first zone.

The lack of smoothness of the curve obtained with model C, especially between points (g) and (h), is caused by an insufficient number of softening points in the first, narrower process zone. With model B and the same 'fine' mesh the response is indeed smoother and the local maxima of the curve are still very well approximated. The length of the first process zone is initially about 2.3 mm and then, after moving over the second initial crack, it reduces to an approximately constant value of about 1.0 mm. The second process zone reaches a size of about 2.3 mm and then remains constant as well.

For model A the 'coarse' mesh can be successfully used to produce a reasonably smooth response, but the local maxima are now significantly underestimated (Figure 21). The length of the first process zone is initially about 8 mm and then, after moving over the second initial crack, it reduces to an approximately constant value of about 3 mm. The length of the fully developed second process zone is approximately 12 mm. Indeed, the same remark made in Section 4.1, about the approximation related to an artificial reduction of the interface strength, still applies here.

Model A was used in Reference [18] in conjunction with the CAL method to perform the same analysis. However, in Reference [18] an additional snap back was found in the unstable

part between points (c) and (f) because the contact between the opposite sides of the second initial crack was not correctly handled.

For the numerical simulations the following solution parameters have been adopted:

$$\gamma = 1.0\text{e} - 7, \quad N_{\text{it}}^{\text{max}} = 20, \quad N_{\text{inc}}^{\text{max}} = 500, \quad I_d = 6, \quad \beta_{\text{sc}} = 0.5$$

$$N_{\text{str}}^{\text{max}} = 25, \quad \beta_{\text{amp}} = 2.0, \quad \Delta\lambda_{\text{max}} = 4.0, \quad N_p^{\text{max}} = 6$$

For the SLS and DLS methods, $N_{\text{ls}} = 20$, $\eta_{\text{max}} = 5.0$ and $\gamma_{\text{ls}} = 0.4$, except for the analyses reported in Tables VII and VIII.

Tables V–IX show a comparison between the numerical performances of the different algorithms considered. In the last columns the final values of the prescribed displacement, u , are reported. They are in bold when the analysis was not successfully terminated either because $N_{\text{str}} = N_{\text{str}}^{\text{max}}$, or because $N_{\text{inc}} = N_{\text{inc}}^{\text{max}}$. In this example not only do the local-control schemes turn out to be always more efficient than the CAL method, but also several differences arise between the NLS, SLS and DLS methods.

Table V reports the results of the analysis conducted with material model A for the four different algorithms. The CAL method does not fail but requires a large number of increment cuts, as well as a large number of increments. The NLS method requires less increments in the earlier part of the analysis, but becomes very unstable when both cracks simultaneously propagate. It fails after reaching the maximum number of increment cuts, $N_{\text{str}}^{\text{max}}$, which was chosen to be very large ($N_{\text{str}}^{\text{max}} = 25$) in order to make sure that failure occurs when convergence is not achieved even for a very small value of ΔI . The use of line searches is then necessary. The SLS method indeed gives much better results, and no increment cuts are required, in sharp contrast with the 165 and 139 cuts occurred for the CAL and NLS methods, respectively. The DLS method gives slightly better results than the SLS, but very close to them. Hence, in this case the SLS is adequate and the DLS method is unnecessary.

The results of the analyses with material model B are reported in Table VI. The CAL method stops at $u = 23.2$ mm after $500 = N_{\text{inc}}^{\text{max}}$ increments. In fact, after a closer look at the R – u curve, it appears that several elastic unloading paths for the whole structure occur. Hence, despite the ability to achieve convergence in all increments with a maximum number of 5 consecutive increment cuts, the method fails to proceed always in the direction which involves increasing delamination (this result is confirmed also for the analysis with material model C). The local-control method without line searches does not have this problem, but it is again unstable, so that failure occurs because of too many increment cuts, at an earlier stage than for material model A. Using line searches, the SLS method successfully terminates the analysis, with 21 increment cuts. However, the DLS method shows a significantly higher stability as it does not require any increment cuts, with the total number of line searches being lower than that used for the SLS method, despite the need of 316 $\delta\lambda$ -line searches. A sensitivity analysis on the tolerance γ_{ls} used for the line searches was then performed in this case, and the results are reported in Tables VII and VIII. Clearly a high degree of instability characterizes the SLS method, which results in big oscillations of all of the results and, especially, in four failures out of eight cases because of too many increment cuts. The DLS method never fails and always requires a small number of increment cuts. Also the dependence on γ_{ls} is smoother and the values 0.3 and 0.4 seem to be optimal although the differences in performances are not significant. Table IX reports the results of the analyses with model C

Table V. Multiple delamination analysis with material model A and the coarse mesh: numerical performance.

	No. of incr.	Tot no. of iter.	No. of incr. cuts	Max. no. of consecutive incr. cuts	No. of η -line searches	No. of $\delta\lambda$ -line searches	Tot. no. of line searches	CPU time (s)	No. incr. with $\delta\lambda$ -line searches	Final u (mm)
CAL	337	3176	165	5	—	—	—	4146	—	36.1
NLS	219	3881	139	25	—	—	—	4088	—	33.6
SLS	110	802	0	0	466	—	466	975	—	36.1
DLS	108	776	0	0	487	15	502	957	2	36.0

Table VI. Multiple delamination analysis with material model B and the fine mesh: numerical performance.

	No. of incr.	Tot no. of iter.	No. of incr. cuts	Max. no. of consecutive incr. cuts	No. of η -line searches	No. of $\delta\lambda$ -line searches	Tot. no. of line searches	CPU time (s)	No. incr. with $\delta\lambda$ -line searches	Final u (mm)
CAL	500	4917	203	5	—	—	—	18 141	—	23.2
NLS	299	3739	111	25	—	—	—	10 933	—	30.7
SLS	271	2250	21	2	1043	—	1043	7408	—	36.1
DLS	210	1524	0	0	416	316	732	5094	20	36.4

Table VII. Multiple delamination analysis with material model B, the fine mesh and the SLS method: sensitivity of the numerical performance on γ_{ls} .

γ_{ls}	No. of incr.	Tot no. of iter.	No. of incr. cuts	Max. no. of consecutive incr. cuts	No. of η -line searches = Tot. no. of line searches	CPU time (s)	Final u (mm)
0.1	261	2338	28	7	2074	8494	36.0
0.2	199	2514	54	25	906	8029	30.8
0.3	206	2688	61	25	897	8521	30.8
0.4	271	2250	21	2	1043	7408	36.1
0.5	229	3017	71	25	1135	9643	30.8
0.6	271	2231	18	3	979	7335	36.2
0.7	258	2042	13	3	567	6458	36.3
0.8	168	2176	47	25	698	6891	27.8

and gives further confirmation that the only algorithm which is always robust is the DLS method.

Figures 22 and 23 relate to the analysis with material model B and $\gamma_{ls} = 0.5$, with the SLS and DLS methods. The grey curves are the result of an analysis with a very small increment size. Figures 22 and 23 have been obtained by starting the analysis from the same converged values, point A in Figure 22 and point B in Figure 23, and then using the SLS or the DLS methods. The two points are on the descending part between points (g) and (h) of Figure 21, not far from point (h). In Figure 22(a) the first two points coincide because no $\delta\lambda$ -line searches are performed in the first two iterations. Then, in the third and the fifth iteration, the DLS method requires two $\delta\lambda$ -line searches (with 18 iterations in total). The increment size is also changed from $\Delta l = 0.411E-02$ to $\Delta l = 0.387E-02$ and the method converges to point B in 7 Newton iterations. Conversely, from the fourth Newton iteration, the SLS method begins to oscillate between two values leading eventually to an increment cut. Point B is very close to the point at which the whole analysis conducted with the SLS method fails because of too many increment cuts. In Figure 22(b), in relation to the same increment of Figure 22(a), the control function used in the DLS method is plotted against the length, ζ , of the equilibrium path, defined by Equation (26). It appears that the point where the DLS method converges is 'critical' in that, although still increasing, the control function shows a big change of the slope. From point B, the analysis is again re-started with the SLS and the DLS methods. Both methods fail at the first 'trial'. After this, the analysis is repeated with the increment length halved and Figure 23(a) shows the iterative points obtained. The first 12 iterations give the same result, as no $\delta\lambda$ -line searches are required in the DLS method. Then, in the 13th and the 15th iteration, the DLS method requires two $\delta\lambda$ -line searches (with 18 iterations in total). The increment size is also changed from $\Delta l = 0.205E-02$ to $\Delta l = 0.187E-03$ and convergence is achieved in 18 Newton iterations at point C. Conversely, the SLS method does not converge and an increment cut is required. In fact, several other increment cuts are needed for the SLS method and the procedure eventually fails. In Figure 23(b) the control function used in the DLS method within the increment of Figure 23(a) is plotted against the arc length, ζ . Although the control function has slightly changed (one out of six points has

Table VIII. Multiple delamination analysis with material model B, the fine mesh and the DLS method: sensitivity of the numerical performance on γ_{ls} .

γ_{ls}	No. of incr.	Tot no. of iter.	No. of incr. cuts	Max. no. of consecutive incr. cuts	No. of η -line searches	No. of $\delta\lambda$ -line searches	Tot. no. of line searches	CPU time (s)	No. incr. with $\delta\lambda$ -line searches	Final u (mm)
0.1	234	1761	7	2	1541	292	1833	6661	18	36.1
0.2	223	1688	4	2	945	263	1208	5950	21	36.1
0.3	210	1513	0	0	486	253	739	5075	12	36.0
0.4	210	1524	0	0	416	316	732	5094	20	36.4
0.5	228	2687	2	1	581	250	831	5653	20	36.1
0.6	229	1755	6	2	621	158	779	5812	16	36.3
0.7	245	1870	4	1	681	330	1011	6322	26	36.3
0.8	214	1607	2	2	377	208	585	5219	15	36.3

Table IX. Multiple delamination analysis with material model C and the fine mesh: numerical performance.

	No. of incr.	Tot no. of iter.	No. of incr. cuts	Max. no. of consecutive incr. cuts	No. of η -line searches	No. of $\delta\lambda$ -line searches	Tot. no. of line searches	CPU time (s)	No. incr. with $\delta\lambda$ -line searches	Final u (mm)
CAL	500	7891	384	5	—	—	—	27942	—	17.2
NLS	122	2151	77	25	—	—	—	6279	—	18.4
SLS	125	2149	61	25	862	—	862	6940	—	22.1
DLS	278	2109	12	4	1140	1301	2441	8187	34	36.0

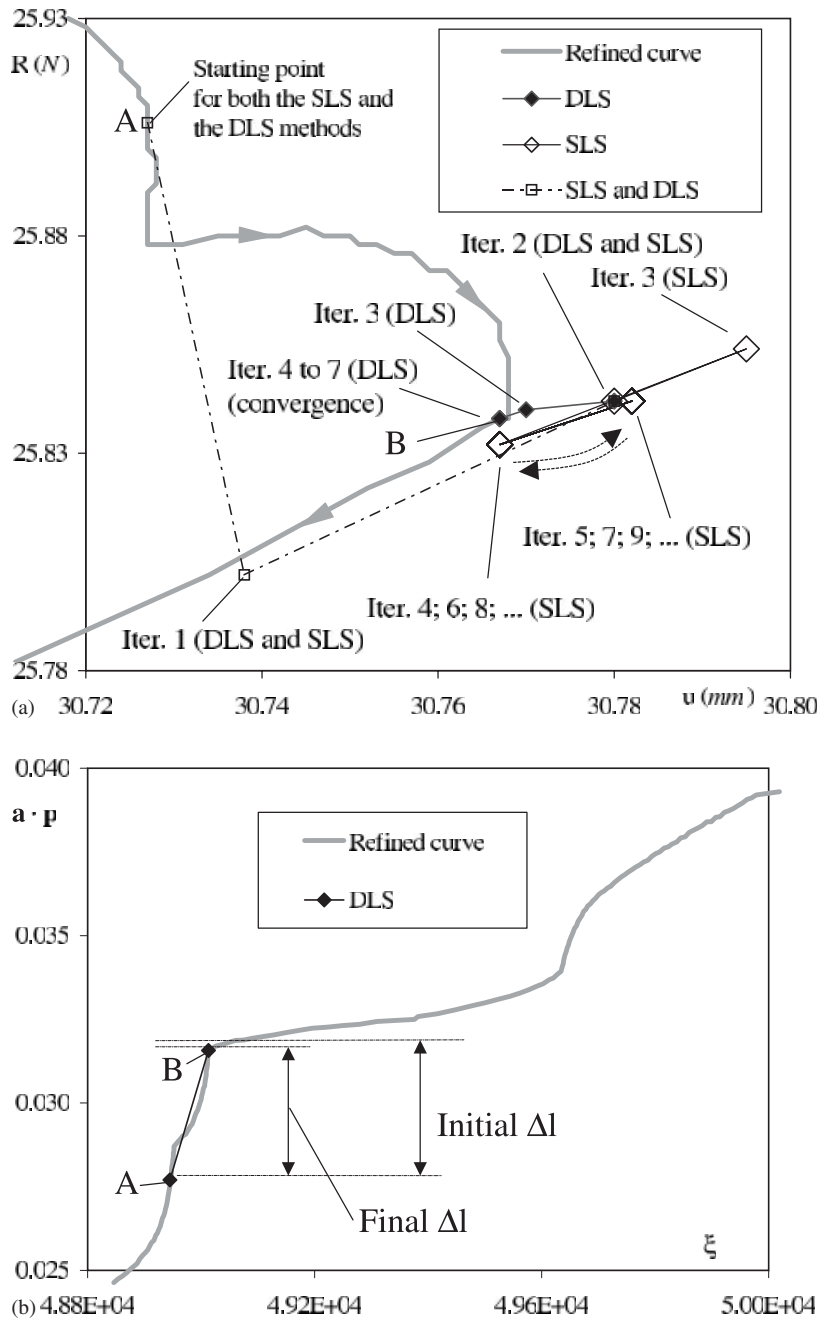


Figure 22. (a) Multiple delamination analysis: iterative values obtained with the SLS and DLS methods; and (b) multiple delamination analysis with the DLS method: control function versus the arc length, ξ , of the equilibrium path.

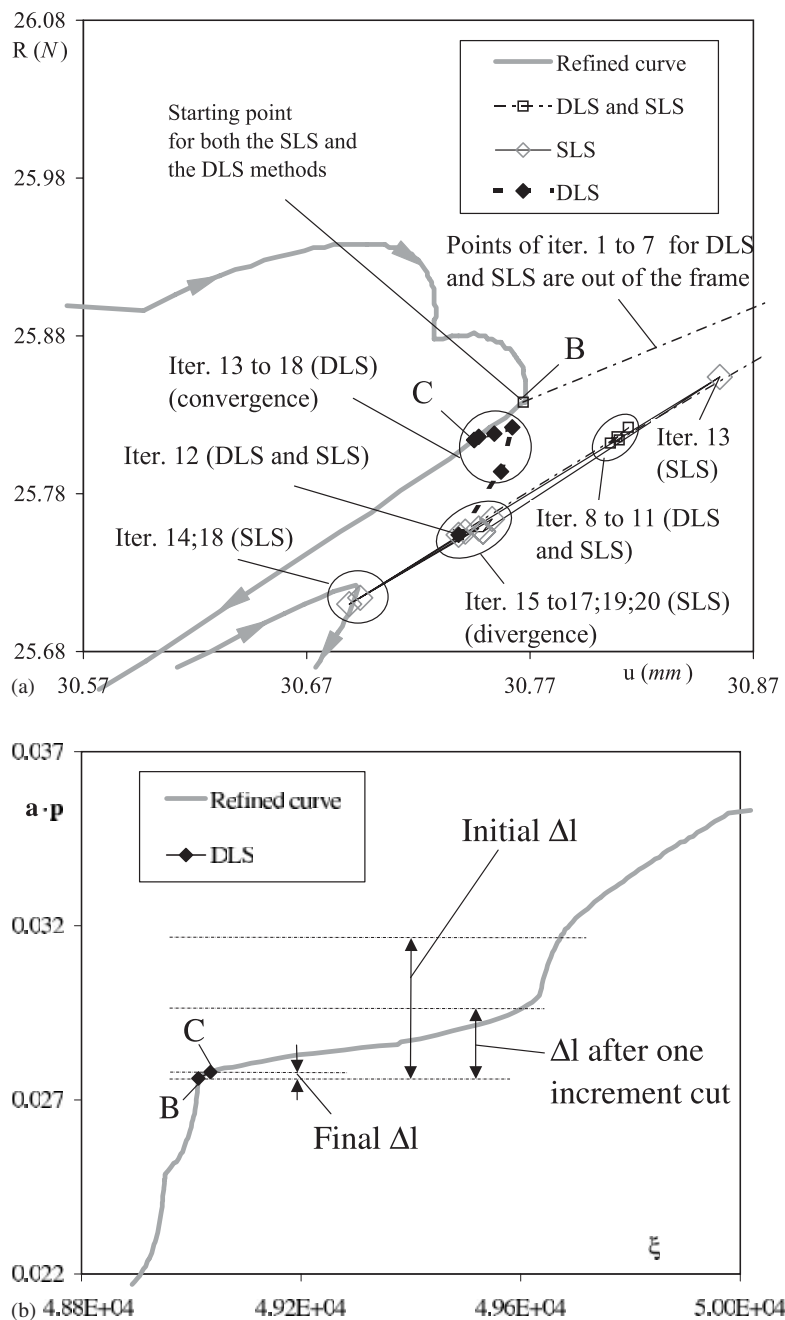


Figure 23. (a) Multiple delamination analysis: iterative values obtained with the SLS and DLS methods (next increment); and (b) multiple delamination analysis with the DLS method: control function versus the arc length, ξ , of the equilibrium path (next increment).

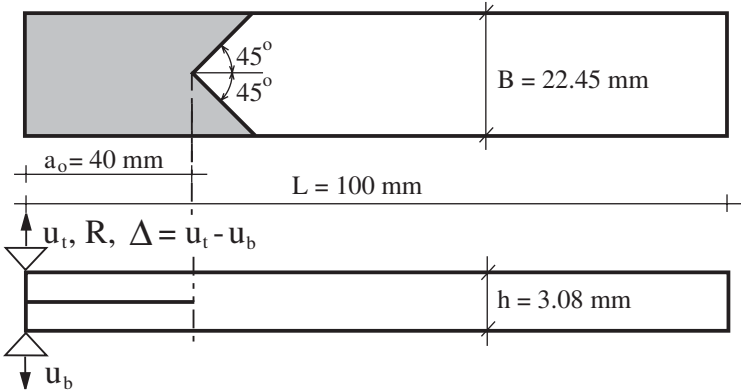


Figure 24. DCB specimen with a V-shaped initial-crack: geometry.

Table X. DCB specimen with a V-shaped initial-crack: material properties [42].

Composite Hexcel T800-924			
E_{11} (GPa)	$E_{22} = E_{33}$ (GPa)	G_{12} (GPa)	$\nu_{12} = \nu_{13} = \nu_{23}$
147.8	9.5	4.6	0.27
Interface			
G_{cI} (N mm ⁻¹)	δ_{oI} (mm)	t_{oI} (MPa)	
Linearly varying between 0.242 at $a = a_0$ and 0.305 at $a = L$	10^{-6}	30.0	

been replaced by an other), it is apparent that the elbow of Figure 22(b) is still present. The DLS method overcomes it using a temporary reduced value of Δl . Then it quickly moves out of the area of Figure 23(a) before encountering a new sharp turn in the equilibrium path, five increments later.

The results discussed in this section, as was explained earlier, have been obtained using displacement control. The same analyses have also been performed using load control, and the numerical performances were similar to those reported in Tables V–IX.

4.3. Three-dimensional analysis of a DCB with a V-shaped initial crack

The analysis of a double cantilever beam with a V-shaped initial crack was performed with a three-dimensional model and displacement control. The geometry and loading are shown in Figure 24. Only mode I is active for symmetry reasons, and the related material properties are given in Table X. They correspond to those reported in Reference [42], except for the tensile strength, which was estimated. Because of the double symmetry of the problem only one half of the top beam was analysed, with the irregular mesh of 420 8-noded thick shell elements shown in Figure 25. The initial undamaged interface was modeled with 400

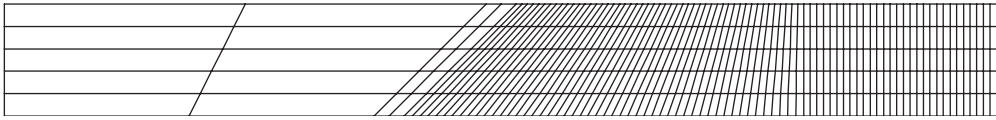
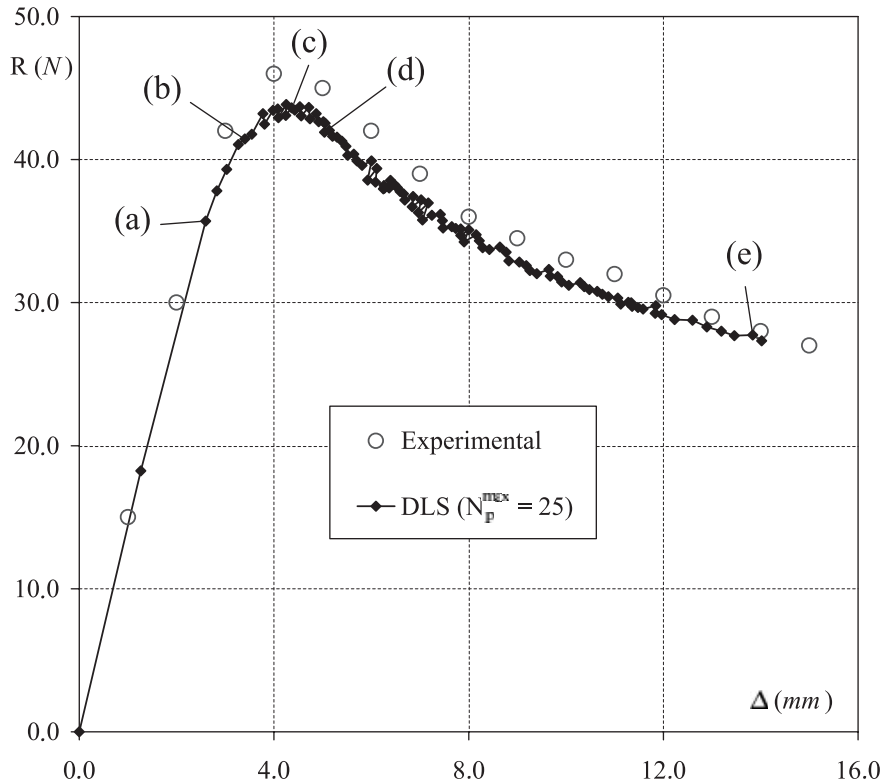


Figure 25. DCB specimen with a V-shaped initial-crack: adopted mesh.

Figure 26. DCB specimen with a V-shaped initial-crack: numerically obtained R - Δ curve and comparison with the experimental results reported in Reference [42].

16-noded interface elements INT16, whereas 20 contact elements INT16C were placed on the V-shaped initial crack. The opening displacement $\Delta = u_t - u_b$ was prescribed and the analysis was considered successfully terminated at increment n if $\Delta_n > 14.0$ mm:

The following solution parameters have been adopted:

$$\gamma = 1.0e-7, \quad N_{it}^{max} = 20, \quad N_{inc}^{max} = 250, \quad I_d = 6, \quad \beta_{sc} = 0.5$$

$$N_{str}^{max} = 25, \quad \beta_{amp} = 2.0, \quad \Delta\lambda_{max} = 4.0$$

For the SLS and DLS methods, the chosen line-search control parameters were $N_{ls} = 20$, $\eta_{max} = 5.0$ and $\gamma_{ls} = 0.4$. The maximum number of controlled relative-displacement components, N_p^{max} , was initially set to 25.

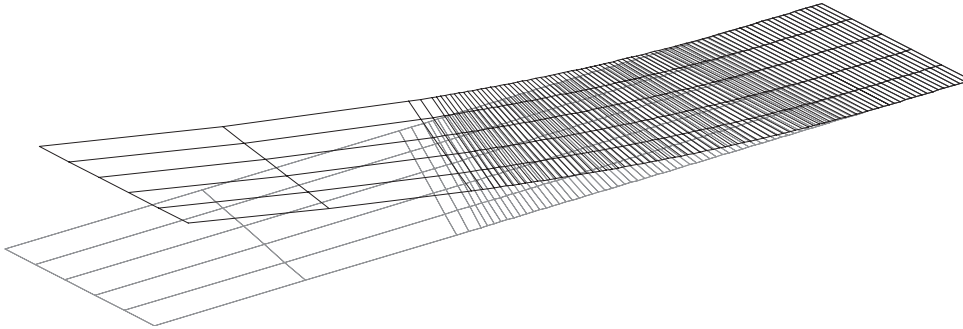


Figure 27. DCB specimen with a V-shaped initial-crack: final deformed mesh.

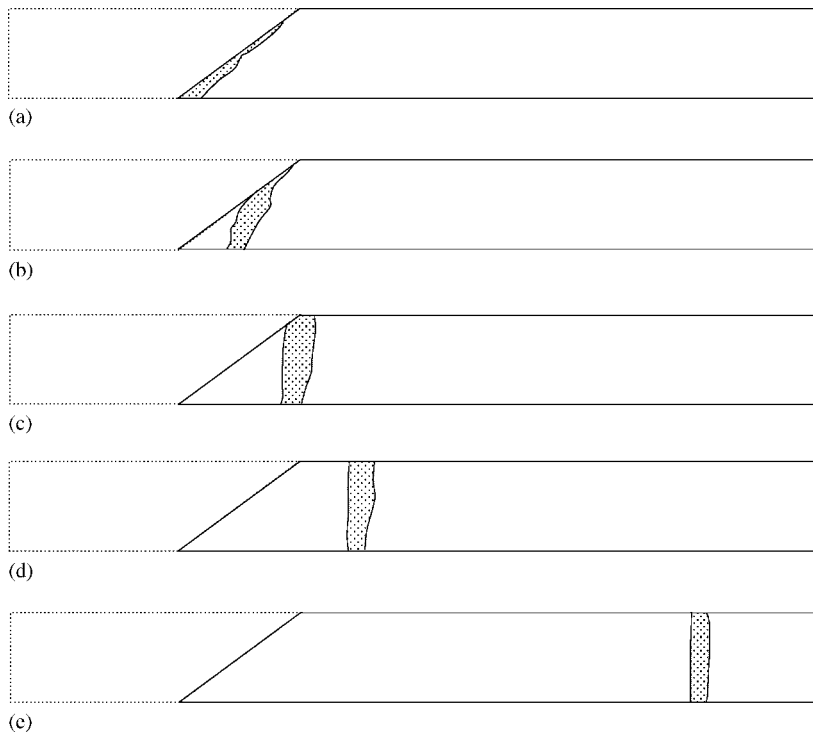


Figure 28. DCB specimen with a V-shaped initial crack: propagation of the process zone.

In Figure 26 the total reaction of the support, R , is plotted against the opening displacement, Δ . The curve obtained with the DLS method is compared with the experimental data reported in Reference [42] and good agreement is found. The final deformed mesh is shown in Figure 27.

Figure 28 shows the propagation of the process zone over the interface in correspondence to the five points marked in Figure 26. The V-shaped initial crack influences the shape of the process zone in the ascending part of the curve, so that the sharp response typically

Table XI. DCB specimen with a V-shaped initial-crack: numerical performance for $N_p^{\max} = 25$.

	No. of incr.	Tot no. of iter.	No. of incr. cuts	Max. no. of consecutive incr. cuts	No. of η -line searches	No. of $\delta\lambda$ -line searches	Tot. no. of line searches	CPU time (s)	No. incr. with $\delta\lambda$ -line searches	Final Δ (mm)
CAL	250	3312	114	4	—	—	—	63 629	—	8.24
NLS	71	542	3	1	—	—	—	8099	—	14.1
SLS	105	803	4	1	651	—	651	14 502	—	14.0
DLS	103	817	4	2	974	246	1220	17 091	8	14.0

Table XII. DCB specimen with a V-shaped initial-crack: numerical performance for $N_p^{\max} = 12$.

	No. of incr.	Tot no. of iter.	No. of incr. cuts	Max. no. of consecutive incr. cuts	No. of η -line searches	No. of $\delta\lambda$ -line searches	Tot. no. of line searches	CPU time (s)	No. incr. with $\delta\lambda$ -line searches	Final Δ (mm)
NLS	122	1180	18	4	—	—	—	17 584	—	14.3
SLS	141	1356	21	3	1073	—	1073	24 503	—	14.0
DLS	111	858	3	1	938	109	1047	16 932	5	14.0

found for a DCB around the peak load is replaced by a rather rounded curve. At point (c) the maximum of the curve is reached and the process zone has almost reached a shape approximately delimited by two straight lines, which is then kept constant in the remaining part of the process, see points (d) and (e) (it would seem that the mesh is not fine enough to show the slight lag in the crack front at the free edge reported in Reference [17]).

In Table XI the numerical performances of the four considered algorithms are compared. Again the CAL method is very inefficient as it requires a lot more increment cuts than are required by any of the local-control methods and eventually fails because it reaches the maximum number of allowed increments.

It is interesting to notice that, for this case, not only is the NLS method adequate to successfully complete the analysis with only 3 increment cuts, but also the total number of increments and of iterations was reduced with respect to the SLS and the DLS method. Considering also the computational cost paid by these last two methods because of the line-search iterations, the NLS method is the best in terms of CPU time. This shows that line searches are not always beneficial to the overall efficiency of the analysis.

The situation changes if a different $N_p^{\max}$, is chosen. As is indicated by Table XII, $N_p^{\max} = 12$ gives a worse performance than $N_p^{\max} = 25$ for the NSL and the SLS methods. Furthermore, the SLS method still gives a worse performance than the NLS method, in terms of number of increments and iterations and, therefore, also in terms of CPU time. Instead, the DLS method gives the best results, even in terms of CPU time.

5. CONCLUSIONS

More robust solution procedures for a class of problems involving a fracturing-softening region have been developed. They have been implemented for the analysis of delamination of laminated composites, but their range of application seems to be easily extendable. The key points are the arc-length method used, which is based on a localized control, and the advanced use of line-search techniques. At each increment of the analysis a linear control function is built up by choosing a user-defined number of controlled relative-displacement parameters within the process zone. If line searches are not used, the local-control method is more efficient than a version of the cylindrical arc-length method recently devised to handle non-smooth problems, but is also more unstable. One strategy, named ‘single-line-search’ (SLS) method and based on the combination of the local-control arc length with ‘conventional’ line-search techniques often, but not always, increases the stability and the efficiency of the algorithm. A new strategy, based on a ‘double-line-search’ (DLS) procedure, has then been developed. When the classical line search fails to compute an optimal step length, with the DLS method a different type of line search is performed in order to modify the iterative change of the load factor so as to obtain a ‘better’ iterative search direction.

A significant improvement has been achieved with the DLS method, which is also the only method to be robust for all of the numerical simulations done. In particular, a significant example involving multiple delamination has shown that, even with an artificially reduced value of the interface strength, the local-control method with no line searches (NLS) fails to converge before the end of the analysis, which is instead completed using the SLS or the DLS methods. For the same example, if the interface strength is increased to more realistic values, the SLS method is also not able to terminate the analysis whereas the DLS method is

always successful. In general, the numerical results reported indicate that, although for some types of problem line searches may be not required at all, for some others it may be at least necessary to introduce conventional line searches through the SLS method, whereas in order to successfully terminate the most difficult analyses the DLS method is needed.

The treatment has been restricted to geometrically linear problems, and future work will consider the extension to geometric non-linearity and to other types of arc-length method. Because of the highly beneficial effect of line searches, methods which reduce their overall computational cost are worthy of investigation. Future work will also involve a comparison of the new developed implicit algorithms with fully explicit strategies, and a combination of the quasi-static approach with dynamics in order to directly follow the dynamic snaps.

APPENDIX A: $\delta\lambda$ -LINE SEARCH FOR CASES b2 AND b3

In this section the $\delta\lambda$ -line search algorithm is detailed for cases b2 and b3 of Section 3.3.2. In the following it is useful to denote by $\text{extrap}(\delta\lambda_h, \delta\lambda_k)$ the value of $\delta\lambda$ obtained by linear extrapolation of $(\delta\lambda_h, \rho_h^1)$ and $(\delta\lambda_k, \rho_k^1)$ (see Figure A1(a)):

$$\text{extrap}(\delta\lambda_h, \delta\lambda_k) = \delta\lambda_h - \frac{\rho_h^1}{\rho_k^1 - \rho_h^1} (\delta\lambda_k - \delta\lambda_h) \quad (\text{A1})$$

Furthermore, given three values $\delta\lambda_h < \delta\lambda_k < \delta\lambda_p$ such that $\rho_k^1 > \rho_h^1$ and $\rho_k^1 > \rho_p^1$, the following ‘quadratic-fitting’ formula [43] is used to compute the value, $\delta\lambda_q$, where the maximum of the quadratic polynomial interpolating $(\delta\lambda_h, \rho_h^1)$, $(\delta\lambda_k, \rho_k^1)$ and $(\delta\lambda_p, \rho_p^1)$ is attained, and we will write:

$$\delta\lambda_q = \text{quadfit}(\delta\lambda_h, \delta\lambda_k, \delta\lambda_p) = \frac{1}{2} \left(\frac{\rho_h^1 b_{kp} + \rho_k^1 b_{ph} + \rho_p^1 b_{hk}}{\rho_h^1 a_{kp} + \rho_k^1 a_{ph} + \rho_p^1 a_{hk}} \right) \quad (\text{A2})$$

where $a_{ij} = \delta\lambda_i - \delta\lambda_j$ and $b_{ij} = \delta\lambda_i^2 - \delta\lambda_j^2$, see Figure A1(f).

The first known value, $\delta\lambda_1$, is such that $\rho_1^1 = \rho^1(\delta\lambda_1) \leq -\gamma_{ls} < 0$, and we also have assumed that $\delta\lambda_1 < \tilde{\delta\lambda}$, with $\tilde{\delta\lambda}$ given by (45). The second value is simply obtained as $\delta\lambda_2 = (\delta\lambda_1 + \tilde{\delta\lambda})/2$.

If $\rho_2^1 = \rho^1(\delta\lambda_2) > -\gamma_{ls}$, we either have convergence or can continue with the interpolation scheme discussed in Section 3.3.2. Otherwise, if $\rho_1^1 \neq \rho_2^1$, we try a linear extrapolation and compute $\delta\lambda_{\text{ext}} = \text{extrap}(\delta\lambda_1, \delta\lambda_2)$. If $\delta\lambda_{\text{ext}} < \tilde{\delta\lambda}$, we take $\delta\lambda_3 = \delta\lambda_{\text{ext}}$ (Figure A1(a)), otherwise we choose the value, $\delta\lambda_3 = 2\delta\lambda_1 - \tilde{\delta\lambda}$ (Figure A1(b)), with a view to ‘screening’ the area $\delta\lambda < \delta\lambda_1$.

For the h th iteration of the algorithm, we proceed as follows. If any value $\delta\lambda_p$ exists such that $\rho_p^1 > -\gamma_{ls}$, we either have convergence or can continue with the interpolation scheme. Otherwise we first reorder the values, $\delta\lambda_p$, so that $\delta\lambda_1 > \delta\lambda_2 > \dots > \delta\lambda_h$. Then, if $\rho_1^1 > \rho_2^1$, and $\text{extrap}(\delta\lambda_1, \delta\lambda_2) < \tilde{\delta\lambda}$, we take $\delta\lambda_{h+1} = \text{extrap}(\delta\lambda_1, \delta\lambda_2)$ (Figure A1(c)). Otherwise, if $\rho_h^1 > \rho_{h-1}^1$, we take $\delta\lambda_{h+1} = \text{extrap}(\delta\lambda_{h-1}, \delta\lambda_h)$ (Figure A1(d)). If none of these things happen, we search for a $p \in \{1, 2, \dots, h-2, h-1\}$ such that $\rho_p^1 \geq \rho_i^1$, for all i . If $p=1$, then we use again the mid-point scheme, and take $\delta\lambda_{h+1} = (\delta\lambda_1 + \tilde{\delta\lambda})/2$ (Figure A1(e)). Otherwise, we use the quadratic-fitting formula (A2) and take $\delta\lambda_{h+1} = \text{quadfit}(\delta\lambda_{p-1}, \delta\lambda_p, \delta\lambda_{p+1})$ (Figure A1(f)).

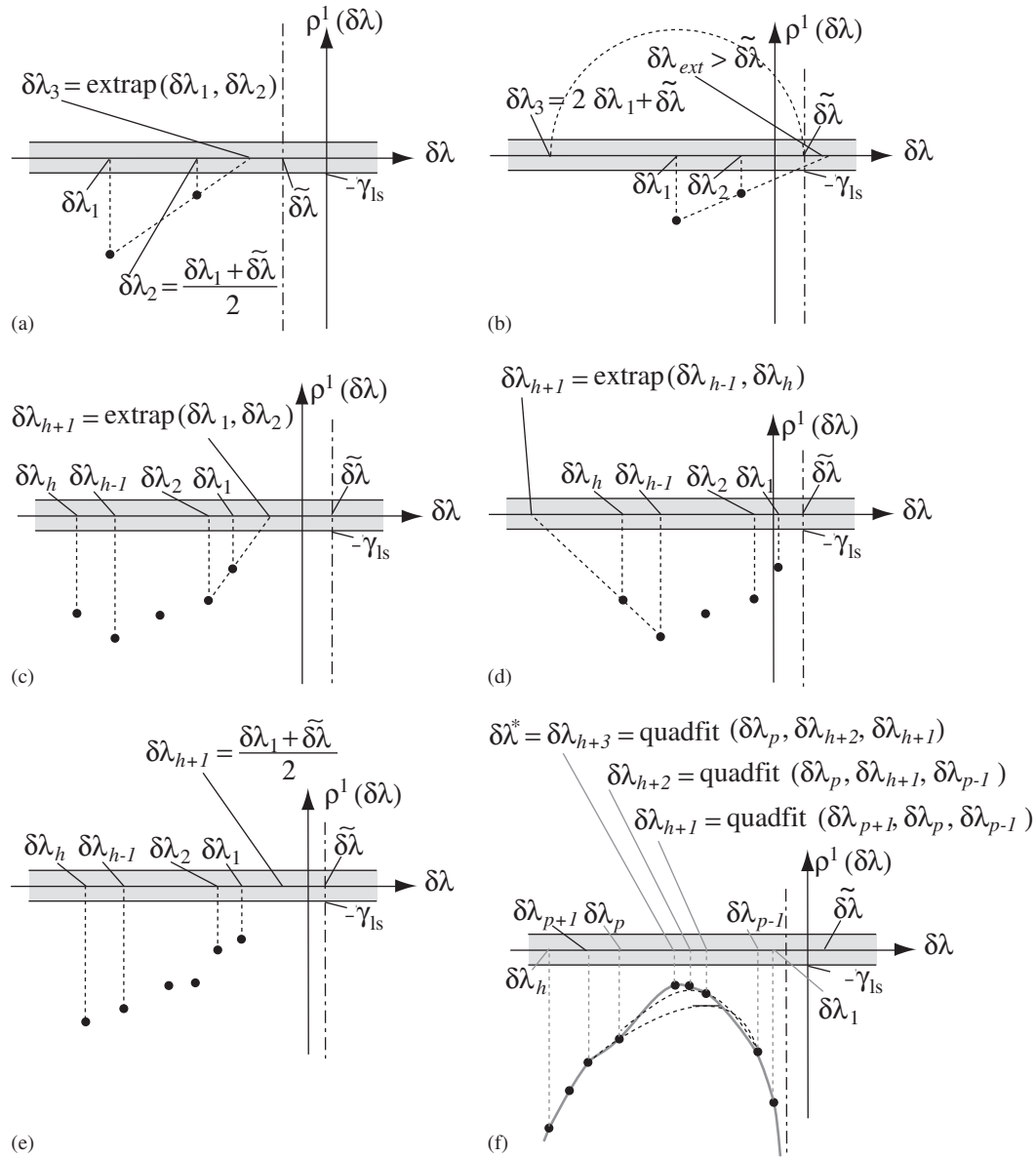


Figure A1. (a) Linear extrapolation; (b) $\delta\lambda_3 = 2\delta\lambda_1 - \tilde{\delta\lambda}$; (c) extrapolation to the right; (d) extrapolation to the left; (e) mid-point scheme; and (f) quadratic-fitting.

After using formula (A2) for 3 times, we stop the $\delta\lambda$ -line search and assume that a good approximation of a (negative) local maximum has been computed. If convergence is not achieved after N_{ls} iterations, we stop the $\delta\lambda$ -line search and take $\delta\lambda^* = \delta\lambda_{N_{ls}}$.

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REFERENCES

1. Dugdale DS. Yielding of steel sheets containing slits. *Journal of the Mechanics and Physics of Solids* 1960; **8**:100–104.
2. Barenblatt GI. The mathematical theory of equilibrium cracks in brittle fracture. *Advances in Applied Mechanics* 1962; **7**:55–129.
3. Hilleborg A, Mod  er M, Petersson PE. Analysis of crack formation and crack growth in concrete by means of fracture mechanics and finite elements. *Cement and Concrete Research* 1976; **6**:773–782.
4. Needleman A. A continuum model for void nucleation by inclusion debonding. *Journal of Applied Mechanics* 1987; **54**:525–531.
5. Needleman A. An analysis of tensile decohesion along an interface. *Journal of the Mechanics and Physics of Solids* 1990; **38**(3):289–324.
6. Tvergaard V, Hutchinson JW. The relation between crack growth resistance and fracture process parameters in elastic-plastic solids. *Journal of the Mechanics and Physics of Solids* 1992; **40**(6):1377–1397.
7. Tvergaard V, Hutchinson JW. The influence of plasticity on mixed mode interface toughness. *Journal of the Mechanics and Physics of Solids* 1993; **41**(6):1119–1135.
8. Wei Y, Hutchinson JW. Interface strength, work of adhesion and plasticity in the peel test. *International Journal of Fracture* 1998; **93**:315–333.
9. Schellekens JCJ, de Borst R. A non-linear finite element approach for the analysis of mode-I free edge delamination in composites. *International Journal of Solids and Structures* 1993; **30**(9):1239–1253.
10. Corigliano A. Formulation, identification and use of interface models in the numerical analysis of composite delamination. *International Journal of Solids and Structures* 1993; **30**(20):2779–2811.
11. Allix O, Lad  veze P, Corigliano A. Damage analysis of interlaminar fracture specimens. *Composite Structures* 1995; **31**(1):61–74.
12. Point N, Sacco E. A delamination model for laminated composites. *International Journal of Solids and Structures* 1996; **33**(4):483–509.
13. Bolzon G, Corigliano A. A discrete formulation for elastic solids with damaging interfaces. *Computer Methods in Applied Mechanics and Engineering* 1997; **140**:329–359.
14. Allix O, Corigliano A. Geometrical and interfacial non-linearities in the analysis of delamination in composites. *International Journal of Solids and Structures* 1999; **36**(15):2189–2216.
15. Mi Y, Crisfield MA, Hellweg HB, Davies GAO. Finite element method and progressive failure modelling of composite structures. In *Computational Plasticity: Fundamentals and Applications. Part I*, Owen DRJ et al. (eds). CIMNE: Barcelona, 1997; 239–254.
16. Chen J, Crisfield MA, Kinloch AJ, Matthews FL, Busso E, Qiu Y. The application of interface elements in predicting progressive delamination of composite material specimens. In *Mechanics of Composite Materials and Structures*, Mota Soares A. et al. (eds), vol. 6, Troia: Portugal, 1998; 1–16.
17. Mi Y, Crisfield MA, Davies GAO, Hellweg HB. Progressive delamination using interface elements. *Journal of Composite Materials* 1998; **32**(14):1246–1272.
18. Alfano G, Crisfield MA. Finite element interface models for the delamination analysis of laminated composites: mechanical and computational issues. *International Journal for Numerical Methods in Engineering* 2001; **50**(7):1701–1736.
19. Riks E, Rankin CC, Brogan FA. On the solution of mode jumping phenomena in thin-walled shell structures. *Computer Methods in Applied Mechanics and Engineering* 1996; **136**(1–2):59–92.
20. de Borst R. Computation of post-bifurcation and post-failure behavior of strain-softening solids. *Computers and Structures* 1987; **25**(2):211–224.
21. Chen Z, Schreyer HL. A numerical solution scheme for softening problems involving total strain control. *Computers and Structures* 1990; **37**(6):1043–1050.
22. May IM, Duan Y. A local arc-length procedure for strain softening. *Computers and Structures* 1997; **64**(1–4):297–303.
23. Geers MGD. Enhanced solution control for physically and geometrically non-linear problems. Part I—the subplane control approach. *International Journal for Numerical Methods in Engineering* 1999; **46**(2):177–204.
24. Hellweg HB, Crisfield MA. A new arc-length method for handling sharp snap-backs. *Computers and Structures* 1998; **66**(5):705–709.

25. Schweizerhof KH, Wriggers P. Consistent linearization for path following methods in nonlinear f.e. analysis. *Computer Methods in Applied Mechanics and Engineering* 1986; **59**:261–279.
26. Crisfield MA. An arc-length method including line searches and accelerations. *International Journal for Numerical Methods in Engineering* 1983; **19**:1269–1289.
27. Crisfield MA. *Non-linear Finite Element Analysis of Solids and Structures*, vol. 1. Wiley: Chichester, 1991.
28. Crisfield MA. *Non-Linear Finite Element Analysis of Solids and Structures*, vol. 2. Wiley: Chichester, 1997.
29. Schweizerhof K. Consistent concept for line search algorithms in combination with arc-length constraints. *Communications in Numerical Methods in Engineering* 1993; **9**(9):773–784.
30. Crisfield MA, Burton AJ. Perspectives of non-linear simulation. In *Benefiting from Thermal and Mechanical Simulation in Micro-Electronics. Proceedings of EuroSimE 2000*, Zhang GQ, Ernst LJ, de Saint Leger O (eds). Boston, Kluwer Academic Publishers: Dordrecht, 2000.
31. Belytschko T, Black T. Elastic crack growth in finite elements with minimal remeshing. *International Journal for Numerical Methods in Engineering* 1999; **45**(5):601–620.
32. Crisfield MA, Alfano G. Adaptive hierarchical enrichment for delamination fracture using a decohesive zone model. *International Journal for Numerical Methods in Engineering* 2002; **54**(9):369–1390.
33. Wu EM, Reuter Jr. RC. Crack extension in fiberglass reinforced plastics and a critical examination of the general fracture criterion. TAM Report 275, University of Illinois, 1965.
34. Reeder JR. An evaluation of mixed-mode delamination failure criteria. Report NASA, TM 104210, 1992.
35. Qiu Y, Crisfield MA, Alfano G. An interface element formulation for the simulation of delamination with buckling. *Engineering Fracture Mechanics* 2001; **68**(16):1755–1776.
36. Crisfield MA. A fast incremental/iterative solution procedure that handles ‘snap-through’. *Computers and Structures* 1981; **13**:55–62.
37. Ramm E. Strategies for tracing the non-linear response near limit points. In *Nonlinear Finite Element Analysis in Structural Mechanics*, Bathe KJ, Stein E, Wunderlich W (eds). Springer: Berlin, 1981; 63–89.
38. Feng YT, Perić D, Owen DRJ. A new criterion for determination of initial loading parameter in arc-length methods. *Computers and Structures* 1996; **58**(3):479–485.
39. Shi J, Crisfield MA. Combining arc-length control and line searches in path following. *Communications in Numerical Methods in Engineering* 1995; **11**(10):793–803.
40. *LUSAS—User Manual*. FEA Ltd: Kingston-upon-Thames, UK, 1999.
41. Robinson P, Song DQ. A modified DCB specimen for mode I testing of multidirectional laminates. *Journal of Composite Materials* 1992; **26**(11):1554–1577.
42. Robinson P, Besant T, Hitchings D. Delamination growth prediction using a finite element approach. In *Fracture of Polymers, Composites and Adhesives*, Williams JG, Pavon A (eds). Elsevier: Amsterdam, 2000, 135–147.
43. Luenberger DG. *Linear and Non-Linear Programming* (2nd edn). Addison-Wesley: Reading, 1984.